



GE VERNOVA

ADMS

16.25.1

Distribution Management System (DMS) Interface Server
Programmer Guide

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Change History

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1. Introduction

This section provides an overview of the ADMS DMS interface server design.

The DMS interface server provides a standard web server interface to allow third-party applications to communicate with the DMS server (NetSrv). The DMS interface server acts as a client to the DMS Server, using the same communication protocol that the Browser client uses to talk to the DMS server. This enables access to the same set of DMS data as the client.

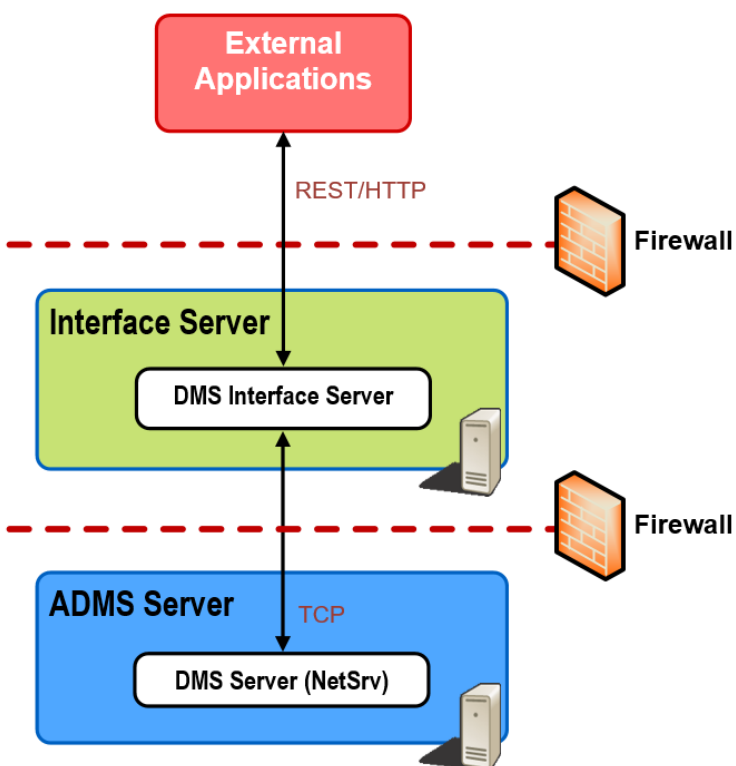


Figure 1. DMS Interface Server Design

Currently the DMS interface server provides the following functionalities:

- **DMS Data Service:** Lists all the available data collections in the system and URLs to access the data such as DNAF solutions, annotations, abnormal topology conditions, and exceptions.
- **DMS Query Service:** Allows external applications to send ad hoc DMS queries to the system. This query interface allows both GET and POST calls.
- **DMS Control Service:** Allows external applications to update DMS data (such as switch statuses and device data), send requests to the Study server, and update annotations. The service is implemented as a set of REST POST calls. Service calls support both XML and JSON formats.
- **Switching Order Web Service:** Allows external applications to update switching orders. This Order web service is implemented as a set of REST POST calls.
- **Annotation Web Service:** Allows showing data from external systems on the network view by creating, modifying, and deleting annotations. This web service is based on the DMS Control

service.

- **Schedules Web Service:** Allows sending or updating schedules. This web service is based on the DMS Control service.

The web interface is implemented using Windows Communication Foundation (WCF). DMS Data Collections specifically is provided through the Open Data Protocol (OData). For details of those standards, refer to <http://msdn.microsoft.com/en-us/data/bb931106> for WCF and <http://www.odata.org> for OData.

1.1. Process Management

The DMS interface server can be managed by CFGMAN and PROCMAN to provide redundancy and high availability. The detailed configuration depends on the deployment needs. For more information, refer to the *Habitat Configuration Manager User Guide* and the *PROCMAN User Guide*.

1.2. Security

The DMS interface server can be run using standard HTTP or HTTPS. To provide additional security for control service calls, an "authorization" header is needed for all POST update requests. The authorization header has the following format:

```
"authorization: DmsISClient:72bPJGtWk0Fuv6lKndo/KA=="
```

For details about how to calculate the hash value for the authorization header, see [Authorization Header](#).

1.3. DMS Journal

The DMS interface server writes distribution management system (DMS) data to the DMS archive journal. The DMS interface server queries data from the DMS server via periodic queries and writes the query responses to the DMS archive journal. The following data types are written to the DMS archive journal.

- Distribution Power Flow (DPF) and Fault Location (FL) DNAF application solutions
- Fault Isolation and Service Restoration (FISR), Automatic Feeder Reconfiguration (AFR), and Load and Volt/VAR Management (LVM) application solutions
- Station and feeder exceptions
- Network model file information

2. DMS Data Service

The DMS Data Service is implemented using Open Data Protocol (OData) to make it easier for other applications to consume the data. OData provides a uniform way to expose, structure, query, and manipulate data using REST practices, and JSON or ATOM syntax to describe the payload. OData also provides a uniform way to represent metadata, allowing applications to know more about the type system, relationships, and structure of the data.

Many client libraries are available for different languages. For sample code, refer to <http://www.odata.org/libraries>.

The DMS Data Service is provided through the following home URI: <http://<YourServerIP>:<Port>/data> (For example, <http://localhost:8801/dms/data>). It lists all the available data collections in the system and its URL to access it.

```
<service xmlns:atom=http://www.w3.org/2005/Atom xmlns=http://www.w3.org/2007/app
xml:base="http://localhost:8801/dms/data/">
  <workspace>
    <atom:title>Default</atom:title>
    <collection href="Abnormals">
      <atom:title>Abnormals</atom:title>
    </collection>
    <collection href="AnnotationTypes">
      <atom:title>AnnotationTypes</atom:title>
    </collection>
    <collection href="Annotations">
      <atom:title>Annotations</atom:title>
    </collection>
    <collection href="DeviceInfos">
      <atom:title>DeviceInfos</atom:title>
    </collection>
    <collection href="DsoSolutions">
      <atom:title>DsoSolutions</atom:title>
    </collection>
    <collection href="DsoPlans">
      <atom:title>DsoPlans</atom:title>
    </collection>
    <collection href="DsoSteps">
      <atom:title>DsoSteps</atom:title>
    </collection>
    <collection href="LvmBusVoltages">
      <atom:title>LvmBusVoltages</atom:title>
    </collection>
    <collection href="PotentialFaults">
      <atom:title>PotentialFaults</atom:title>
    </collection>
    <collection href="FaultLocations">
      <atom:title>FaultLocations</atom:title>
    </collection>
    <collection href="FaultLocationSolutions">
      <atom:title>FaultLocationSolutions</atom:title>
    </collection>
  </workspace>
</service>
```



```

</collection>
<collection href="FaultLocationResults">
  <atom:title>FaultLocationResults</atom:title>
</collection>
<collection href="Stations">
  <atom:title>Stations</atom:title>
</collection>
<collection href="StationFiles">
  <atom:title>StationFiles</atom:title>
</collection>
<collection href="DpfSolutions">
  <atom:title>DpfSolutions</atom:title>
</collection>
<collection href="StationExceptions">
  <atom:title>StationExceptions</atom:title>
</collection>
<collection href="FeederExceptions">
  <atom:title>FeederExceptions</atom:title>
</collection>
<collection href="VoltageViolations">
  <atom:title>VoltageViolations</atom:title>
</collection>
<collection href="OverloadViolations">
  <atom:title>OverloadViolations</atom:title>
</collection>
<collection href="VoltageOutputs">
  <atom:title>VoltageOutputs</atom:title>
</collection>
<collection href="MeasurementIslands">
  <atom:title>MeasurementIslands</atom:title>
</collection>
<collection href="DpfFeederLosses">
  <atom:title>DpfFeederLosses</atom:title>
</collection>
<collection href="DpfTfLosses">
  <atom:title>DpfTfLosses</atom:title>
</collection>
<collection href="DsoFeederParameters">
  <atom:title>DsoFeederParameters</atom:title>
</collection>
<collection href="DsoTfParameters">
  <atom:title>DsoTfParameters</atom:title>
</collection>
</workspace>
</service>

```

Navigating to the URI `http://localhost:8801/dms/data/$metadata` will show all the properties of all the collection data in the system.

For each type of collection, you can also include an additional expand/filter to either expand the data or narrow the result. For details about how to construct URLs to get specific data, refer to <https://www.odata.org/documentation/odata-version-2-0/uri-conventions/>.

The following is a list of sample URLs to get specific data:

- Lists all the DSO solutions in the system:

```
http://localhost:8801/dms/data/DsoSolutions
```

- Lists the details of one DSO solution by its key:

```
http://localhost:8801/dms/data/DsoSolutions('Solution_BAKER_1357257300898')
```

- Lists all the plans for this solution:

```
http://localhost:8801/dms/data/DsoSolutions('Solution_BAKER_1357257300898')/DsoPlanList
```

- Lists all the solutions that have plans:

```
http://localhost:8801/dms/data/DsoSolutions?$filter=NumberOfPlans%20gt%200
```

- Lists the solution details with DsoPlanList expanded inline:

```
http://localhost:8801/dms/data/DsoSolutions?$expand=DsoPlanList
```

- Lists all current station information with ModelFileList expanded for each station:

```
http://localhost:8801/dms/data/Stations?$expand=ModelFileList
```

- Lists current station model file information for 3LIONS station:

```
http://localhost:8801/dms/data/StationFiles?$filter=StationName%20eq%20'3LIONS'
```

- Lists current station exceptions for 3LIONS station:

```
http://localhost:8801/dms/data/StationExceptions?$filter=StationName%20eq%20'3LIONS'
```

- Lists current feeder exceptions for 3LIONS station:

```
http://localhost:8801/dms/data/FeederExceptions?$filter=StationName%20eq%20'3LIONS'
```

- Lists all annotation types supported in the system:

```
http://localhost:8801/dms/data/AnnotationTypes
```

- Lists all current annotations in the system:

```
http://localhost:8801/dms/data/Annotations
```

- Lists all current annotations with Type 0 ('KeyCustomer') for RAVEN station:

```
http://localhost:8801/dms/data/Annotations?$filter=Type%20eq%200&StationName%20eq%20'RAVEN'
```

- Lists current DPF solution for 3LIONS station:

```
http://localhost:8801/dms/data/DpfSolutions?$filter=StationName%20eq%20'3LIONS'
```

- Queries a DPF solution with DpfRequestId = 16bc18af-2581-4171-8011-68abad50141f:

```
http://localhost:8801/dms/data/DpfSolutions?$filter=RequestId%20eq%20'16bc18af-2581-4171-8011-68abad50141f'
```

The DMS interface server also provides the following URLs to get specific types of data without using filters.

- Lists all the FISR solutions:

```
http://localhost:8801/dms/data/FISR Solutions
```

- Lists all the VVC solutions:

```
http://localhost:8801/dms/data/VVC Solutions
```

- Lists all the AFR solutions:

```
http://localhost:8801/dms/data/AFR Solutions
```

- Lists all current annotations with "KeyCustomer" type in the system:

```
http://localhost:8801/dms/data/GetAnnotation?TypeName='KeyCustomer'
```

2.1. Entity Types

The URL `http://localhost:8801/dms/data/$metadata` lists all the entity definitions for each data type. For reference, these are listed in the table below.

Name	Entity Type
Abnormal Component	<pre><EntityType Name="AbnormalComponent"> <Key> <PropertyRef Name="Id"/> </Key> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="Name" Type="Edm.String"/> <Property Name="AbnormalTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="FeederName" Type="Edm.String"/> <Property Name="XCoordinate" Nullable="false" Type="Edm.Single"/> <Property Name="YCoordinate" Nullable="false" Type="Edm.Single"/> </EntityType></pre>

Name	Entity Type
Annotation Type	<pre> -<EntityType Name="AnnotationType"> -<Key> <PropertyRef Name="Type"/> </Key> <Property Name="Type" Nullable="false" Type="Edm.Int32"/> <Property Name="Name" Type="Edm.String"/> </EntityType> </pre>
Annotation	<pre> -<EntityType Name="Annotation"> -<Key> <PropertyRef Name="Id"/> </Key> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="Name" Type="Edm.String"/> <Property Name="Type" Nullable="false" Type="Edm.Int32"/> <Property Name="Status" Nullable="false" Type="Edm.Int32"/> <Property Name="StatusText" Type="Edm.String"/> <Property Name="XCoordinate" Nullable="false" Type="Edm.Single"/> <Property Name="YCoordinate" Nullable="false" Type="Edm.Single"/> <Property Name="Latitude" Nullable="false" Type="Edm.Double"/> <Property Name="Longitude" Nullable="false" Type="Edm.Double"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="ServiceCenter" Type="Edm.String"/> <Property Name="DeviceId" = Type="Edm.String"/> <Property Name="DeviceType" = Type="Edm.String"/> <Property Name="CreationTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="UpdateTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="CreatedBy" Type="Edm.String"/> <Property Name="UpdatedBy" Type="Edm.String"/> <Property Name="Comment" Type="Edm.String"/> <Property Name="Comment2" Type="Edm.String"/> <Property Name="Comment3" Type="Edm.String"/> <Property Name="Comment4" Type="Edm.String"/> <Property Name="Comment5" Type="Edm.String"/> </EntityType> </pre>
DeviceInfo	<pre> -<EntityType Name="DeviceInfo"> -<Key> <PropertyRef Name="DeviceId"/> <PropertyRef Name="StationName"/> </Key> <Property Name="DeviceId" Nullable="false" Type="Edm.String"/> <Property Name="DeviceName" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="NominalFeederId" Type="Edm.String"/> <Property Name="NominalFeederName" Type="Edm.String"/> <Property Name="DeviceTypeCode" Type="Edm.Int32"/> <Property Name="DeviceTypeName" Type="Edm.String"/> <Property Name="SwitchTypeCode" Type="Edm.Int32"/> <Property Name="SwitchTypeName" Type="Edm.String"/> <Property Name="OpCenterId" Type="Edm.String"/> <Property Name="CtrlPntCnt" Type="Edm.Int32"/> <Property Name="IsScada" Type="Edm.Boolean"/> </EntityType> </pre>

Name	Entity Type
DsoSolution	<pre> -<EntityType Name="DsoSolution"> -<Key> <PropertyRef Name="SolutionId"/> </Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="NumberOfPlans" Type="Edm.Int32"/> <Property Name="EvaluationTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="PeakEvaluationTime" Type="Edm.DateTime"/> <Property Name="ExecutionTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="ExecutionTimeTicks" Nullable="false" Type="Edm.Int64"/> <Property Name="CompletionTime" Type="Edm.DateTime"/> <Property Name="ProblemFormulation" Type="Edm.String"/> <Property Name="SolutionTypeCode" Nullable="false" Type="Edm.Int32"/> <Property Name="SolutionType" Type="Edm.String"/> <Property Name="DnafParamsVersion" Type="Edm.String"/> <Property Name="DnafParamsLastModifiedTime" Type="Edm.DateTime"/> <Property Name="DnafParamsRevisionNumber" Type="Edm.Int32"/> <Property Name="FeederList" Type="Edm.String"/> <Property Name="DistributionAutomationEnabledFeederList" Type="Edm.String"/> <Property Name="DistributionAutomationEligibleSwitchList" Type="Edm.String"/> <Property Name="IsArchived" Nullable="false" Type="Edm.Boolean"/> <NavigationProperty Name="DsoPlanList" FromRole="DsoSolution" ToRole="DsoPlanList" Relationship="DmsDataModel.DsoSolution_DsoPlanList"/> <NavigationProperty Name="TfParameters" FromRole="DsoSolution" ToRole="TfParameters" Relationship="DmsDataModel.DsoSolution_TfParameters"/> <NavigationProperty Name="FeederParameters" FromRole="DsoSolution" ToRole="FeederParameters" Relationship="DmsDataModel.DsoSolution_FeederParameters"/> </EntityType> </pre>

Name	Entity Type
DsoPlan	<pre> <EntityType Name="DsoPlan"> <Key> <PropertyRef Name="PlanId"/> </Key> <Property Name="PlanId" Nullable="false" Type="Edm.String"/> <Property Name="SolutionId" Type="Edm.String"/> <Property Name="PlanTypeCode" Nullable="false" Type="Edm.Int32"/> <Property Name="PlanType" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="Rank" Type="Edm.Int32"/> <Property Name="State" Type="Edm.String"/> <Property Name="ProtectionValidation" Type="Edm.String"/> <Property Name="FaultComponentId" Type="Edm.String"/> <Property Name="ProblemFormulation" Type="Edm.String"/> <Property Name="VoltageLimitsUsed" Type="Edm.String"/> <Property Name="FlowLimitsUsed" Type="Edm.String"/> <Property Name="RestoredKw" Type="Edm.Double"/> <Property Name="UnservicedKw" Type="Edm.Double"/> <Property Name="LostRevenue" Type="Edm.Double"/> <Property Name="PreplanRevenue" Type="Edm.Double"/> <Property Name="PostplanRevenue" Type="Edm.Double"/> <Property Name="PreplanCustomerValueUninterruptedService" Type="Edm.Double"/> <Property Name="PostplanCustomerValueUninterruptedService" Type="Edm.Double"/> <Property Name="PreplanPowerLoss" Type="Edm.Double"/> <Property Name="PostplanPowerLoss" Type="Edm.Double"/> <Property Name="NumberOfMoves" Type="Edm.Int32"/> <Property Name="NumberOfStations" Type="Edm.Int32"/> <Property Name="NumberOfCustomersDeenergized" Type="Edm.Int32"/> <Property Name="NumberOfCustomersRestored" Type="Edm.Int32"/> <Property Name="NumberOfCustomersNotRestored" Type="Edm.Int32"/> <Property Name="NumberOfHigherPriorityCustomersNotRestored" Type="Edm.Int32"/> <Property Name="NumberOfTransformersDeenergized" Type="Edm.Int32"/> <Property Name="NumberOfTransformersRestored" Type="Edm.Int32"/> <Property Name="NumberOfTransformersNotRestored" Type="Edm.Int32"/> <Property Name="NumberOfVoltageViolationComponents" Type="Edm.Int32"/> <Property Name="MaximumSegmentLoadingPercent" Type="Edm.Double"/> <Property Name="MaximumLoadingSegmentId" Type="Edm.String"/> <Property Name="MaximumLoadVoltage" Type="Edm.Double"/> <Property Name="MaximumVoltageLoadId" Type="Edm.String"/> <Property Name="MinimumLoadVoltage" Type="Edm.Double"/> <Property Name="MinimumVoltageLoadId" Type="Edm.String"/> <Property Name="PreplanWorstVoltageOverload" Type="Edm.Double"/> <Property Name="PostplanWorstVoltageOverload" Type="Edm.Double"/> <Property Name="PreplanVoltageViolationPerformanceIndex" Type="Edm.Double"/> <Property Name="PostplanVoltageViolationPerformanceIndex" Type="Edm.Double"/> <Property Name="PreplanActiveDemand" Type="Edm.Double"/> <Property Name="PostplanActiveDemand" Type="Edm.Double"/> <Property Name="ActiveDemandReduction" Type="Edm.Double"/> <Property Name="PreplanReactiveDemand" Type="Edm.Double"/> <Property Name="PostplanReactiveDemand" Type="Edm.Double"/> <Property Name="ReactiveDemandReduction" Type="Edm.Double"/> <Property Name="MinimumTargetPowerFactor" Type="Edm.Double"/> <Property Name="MaximumTargetPowerFactor" Type="Edm.Double"/> <Property Name="FeederTargetPowerFactor" Type="Edm.Double"/> <Property Name="PreplanFeederPowerFactor" Type="Edm.Double"/> <Property Name="PostplanFeederPowerFactor" Type="Edm.Double"/> <Property Name="PreplanAreaPowerFactor" Type="Edm.Double"/> <Property Name="PostplanAreaPowerFactor" Type="Edm.Double"/> <Property Name="PreplanPowerTransformerPowerFactors" Type="Edm.String"/> <Property Name="PostplanPowerTransformerPowerFactors" Type="Edm.String"/> </pre>

Name	Entity Type
DsoPlan (Cont...)	<pre> <Property Name="IsPostplanPowerFactorWithinLimit" Type="Edm.Boolean"/> <Property Name="IsPostplanNetworkRadial" Type="Edm.Boolean"/> <Property Name="SecondRatingViolationsExist" Type="Edm.Boolean"/> <Property Name="NumberOfCommonBusses" Type="Edm.Int32"/> <Property Name="PreplanCommonBusStandardDeviation" Type="Edm.Double"/> <Property Name="PostplanCommonBusStandardDeviation" Type="Edm.Double"/> <Property Name="PreplanMaifi" Type="Edm.Double"/> <Property Name="PostplanMaifi" Type="Edm.Double"/> <Property Name="PreplanSaidi" Type="Edm.Double"/> <Property Name="PostplanSaidi" Type="Edm.Double"/> <Property Name="PreplanSaifi" Type="Edm.Double"/> <Property Name="PostplanSaifi" Type="Edm.Double"/> <Property Name="TotalPerformanceIndex" Type="Edm.Double"/> <Property Name="NumberOfMovesPerformanceIndex" Type="Edm.Double"/> <Property Name="CostPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateCostPerformanceIndex" Type="Edm.Double"/> <Property Name="FlowViolationPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateFlowViolationPerformanceIndex" Type="Edm.Double"/> <Property Name="VoltageViolationPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateVoltageViolationPerformanceIndex" Type="Edm.Double"/> <Property Name="LostLoadPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateLostLoadPerformanceIndex" Type="Edm.Double"/> <Property Name="LossesPerformanceIndex" Type="Edm.Double"/> <Property Name="DroppedCustomerCountPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateDroppedCustomerCountPerformanceIndex" Type="Edm.Double"/> <Property Name="DroppedCustomerPriorityPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateDroppedCustomerPriorityPerformanceIndex" Type="Edm.Double"/> <Property Name="CmiPerformanceIndex" Type="Edm.Double"/> <Property Name="PriorityCmiPerformanceIndex" Type="Edm.Double"/> <Property Name="LmiPerformanceIndex" Type="Edm.Double"/> <Property Name="OutageEnergizedPerformanceIndex" Type="Edm.Double"/> <Property Name="IsolationDropPoisonLoadPerformanceIndex" Type="Edm.Double"/> <Property Name="MarginPerformanceIndex" Type="Edm.Double"/> <Property Name="RadialPerformanceIndex" Type="Edm.Double"/> <Property Name="IntermediateRadialPerformanceIndex" Type="Edm.Double"/> <Property Name="MaifiPerformanceIndex" Type="Edm.Double"/> <Property Name="SaidiPerformanceIndex" Type="Edm.Double"/> <Property Name="SaifiPerformanceIndex" Type="Edm.Double"/> <Property Name="FeederVoltageDropPerformanceIndex" Type="Edm.Double"/> <Property Name="TculRangePerformanceIndex" Type="Edm.Double"/> <Property Name="ProtectionValidationPerformanceIndex" Type="Edm.Double"/> <Property Name="IsArchived" Nullable="false" Type="Edm.Boolean"/> <NavigationProperty Name="DsoStepList" FromRole="DsoPlan" ToRole="DsoStepList" Relationship="DmsDataModel.DsoPlan_DsoStepList"/> <NavigationProperty Name="BusVoltageList" FromRole="DsoPlan" ToRole="BusVoltageList" Relationship="DmsDataModel.DsoPlan_BusVoltageList"/> </EntityType> </pre>

Name	Entity Type
DsoStep	<pre> -<EntityType Name="DsoStep"> -<Key> <PropertyRef Name="PlanId"/> <PropertyRef Name="StepNumber"/> </Key> <Property Name="PlanId" Nullable="false" Type="Edm.String"/> <Property Name="StepNumber" Nullable="false" Type="Edm.Int32"/> <Property Name="ControlTypeName" Type="Edm.String"/> <Property Name="SwitchingTypeCode" Type="Edm.Int32"/> <Property Name="SwitchingTypeName" Type="Edm.String"/> <Property Name="PhasesCode" Nullable="false" Type="Edm.Int32"/> <Property Name="Phases" Type="Edm.String"/> <Property Name="DeviceId" Type="Edm.String"/> <Property Name="DeviceName" Type="Edm.String"/> <Property Name="DeviceTypeName" Type="Edm.String"/> <Property Name="DeviceTypeMap" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="NominalFeederName" Type="Edm.String"/> <Property Name="InitialState" Type="Edm.String"/> <Property Name="RecommendedState" Type="Edm.String"/> <NavigationProperty Name="DeviceInfo" FromRole="DsoStep" ToRole="DeviceInfo" Relationship="DmsDataModel.DsoStep_DeviceInfo"/> </EntityType> </pre>
Bus Voltage	<pre> -<EntityType Name="BusVoltage"> -<Key> <PropertyRef Name="DeviceId"/> <PropertyRef Name="PlanId"/> </Key> <Property Name="PlanId" Nullable="false" Type="Edm.String"/> <Property Name="DeviceId" Nullable="false" Type="Edm.String"/> <Property Name="DeviceName" Type="Edm.String"/> <Property Name="NominalFeederId" Type="Edm.String"/> <Property Name="NominalFeederName" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="OpCenterId" Type="Edm.String"/> <Property Name="IsScada" Type="Edm.Boolean"/> <Property Name="DynamicFeederName" Type="Edm.String"/> <Property Name="PostplanPhaseAVoltage" Type="Edm.String"/> <Property Name="PostplanPhaseBVoltage" Type="Edm.String"/> <Property Name="PostplanPhaseCVoltage" Type="Edm.String"/> <Property Name="HighVoltageLimit" Type="Edm.String"/> <Property Name="OriginalHighVoltageLimit" Type="Edm.String"/> <Property Name="ManualHighVoltageLimit" Type="Edm.String"/> <Property Name="HighVoltageLimitSource" Type="Edm.String"/> <Property Name="LowVoltageLimit" Type="Edm.String"/> <Property Name="OriginalLowVoltageLimit" Type="Edm.String"/> <Property Name="ManualLowVoltageLimit" Type="Edm.String"/> <Property Name="LowVoltageLimitSource" Type="Edm.String"/> </EntityType> </pre>

Name	Entity Type
Potential Fault Location	<pre> -<EntityType Name="PotentialFaultLocation"> -<Key> <PropertyRef Name="Id"/> </Key> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="PhaseACurrent" Nullable="false" Type="Edm.Single"/> <Property Name="PhaseBCurrent" Nullable="false" Type="Edm.Single"/> <Property Name="PhaseCCurrent" Nullable="false" Type="Edm.Single"/> <Property Name="XCoordinate" Nullable="false" Type="Edm.Single"/> <Property Name="YCoordinate" Nullable="false" Type="Edm.Single"/> </EntityType> </pre>
Fault Location	<pre> -<EntityType Name="FaultLocation"> -<Key> <PropertyRef Name="Id"/> </Key> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="Name" Type="Edm.String"/> <Property Name="AbnDateTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="FeederName" Type="Edm.String"/> <Property Name="Description" Type="Edm.String"/> <Property Name="XCoordinate" Nullable="false" Type="Edm.Single"/> <Property Name="YCoordinate" Nullable="false" Type="Edm.Single"/> </EntityType> </pre>
Fault Location Solution	<pre> -<EntityType Name="FaultLocationSolution"> -<Key><PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="SolutionTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="SolutionTimeTicks" Nullable="false" Type="Edm.Int64"/> <Property Name="DeviceId" Type="Edm.String"/> <Property Name="DeviceName" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="PhaseAFaultCurrent" Type="Edm.Double"/> <Property Name="PhaseBFaultCurrent" Type="Edm.Double"/> <Property Name="PhaseCFaultCurrent" Type="Edm.Double"/> <Property Name="GroundFaultCurrent" Type="Edm.Double"/> <Property Name="PeakFaultCurrent" Type="Edm.Double"/> <Property Name="FaultType" Type="Edm.String"/> <Property Name="NumberOfFaultLocations" Nullable="false" Type="Edm.Int32"/> <NavigationProperty Name="DeviceInfo" FromRole="FaultLocationSolution" ToRole="DeviceInfo" Relationship="DmsDataModel.FaultLocationSolution_DeviceInfo"/> <NavigationProperty Name="ResultList" FromRole="FaultLocationSolution" ToRole="ResultList" Relationship="DmsDataModel.FaultLocationSolution_ResultList"/> </EntityType> </pre>

Name	Entity Type
Fault Location Result	<pre> <EntityType Name="FaultLocationResult"> <Key><PropertyRef Name="DeviceId"/> <PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="PhaseACalculatedFaultCurrent" Type="Edm.Double"/> <Property Name="PhaseACalculatedFaultCurrentDifferencePercent" Type="Edm.Double"/> <Property Name="PhaseBCalculatedFaultCurrent" Type="Edm.Double"/> <Property Name="PhaseBCalculatedFaultCurrentDifferencePercent" Type="Edm.Double"/> <Property Name="PhaseCCalculatedFaultCurrent" Type="Edm.Double"/> <Property Name="PhaseCCalculatedFaultCurrentDifferencePercent" Type="Edm.Double"/> <Property Name="DeviceId" Nullable="false" Type="Edm.String"/> <Property Name="DeviceType" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <NavigationProperty Name="DeviceInfo" FromRole="FaultLocationResult" ToRole="DeviceInfo" Relationship="DmsDataModel.FaultLocationResult_DeviceInfo"/> </EntityType> </pre>
Station	<pre> <EntityType Name="Station"> <Key><PropertyRef Name="Name"/></Key> <Property Name="Id" Type="Edm.String"/> <Property Name="Name" Nullable="false" Type="Edm.String"/> <Property Name="Crc" Type="Edm.String"/> <Property Name="DnafCapable" Type="Edm.String"/> <Property Name="LvmProblemFormulation" Type="Edm.String"/> <Property Name="FisrProblemFormulation" Type="Edm.String"/> <Property Name="AfrProblemFormulation" Type="Edm.String"/> <Property Name="DpfSolutionTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="DpfExecutionTime" Nullable="false" Type="Edm.DateTime"/> <NavigationProperty Name="ModelFileList" FromRole="Station" ToRole="ModelFileList" Relationship="DmsDataModel.Station_ModelFileList"/> <NavigationProperty Name="LatestDpfSolution" FromRole="Station" ToRole="LatestDpfSolution" Relationship="DmsDataModel.Station_LatestDpfSolution"/> </EntityType> </pre>
Station File Info	<pre> <EntityType Name="StationFileInfo"> <Key><PropertyRef Name="Name"/> <PropertyRef Name="StationName"/></Key> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="Name" Nullable="false" Type="Edm.String"/> <Property Name="DirectoryName" Type="Edm.String"/> <Property Name="CreationTime" Type="Edm.DateTime"/> <Property Name="LastWriteTime" Type="Edm.DateTime"/> <Property Name="Length" Type="Edm.Int64"/> <Property Name="ConverterVersion" Type="Edm.String"/> </EntityType> </pre>

Name	Entity Type
DpfSolution	<pre> <EntityType Name="DpfSolution"> <Key><PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="Iterations" Type="Edm.Int32"/> <Property Name="Converged" Type="Edm.Boolean"/> <Property Name="ReferenceBus" Type="Edm.String"/> <Property Name="SolutionReason" Type="Edm.String"/> <Property Name="RequestId" Type="Edm.String"/> <Property Name="SolutionTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="ExecutionTime" Nullable="false" Type="Edm.DateTime"/> <Property Name="DnafParamsVersion" Type="Edm.String"/> <Property Name="DnafParamsLastModifiedTime" Type="Edm.DateTime"/> <Property Name="DnafParamsRevisionNumber" Type="Edm.Int32"/> <Property Name="RealPowerLoss" Type="Edm.Double"/> <Property Name="ReactivePowerLoss" Type="Edm.Double"/> <Property Name="IsArchived" Nullable="false" Type="Edm.Boolean"/> <NavigationProperty Name="stationException" FromRole="DpfSolution" ToRole="stationException" Relationship="DmsDataModel.DpfSolution_stationException"/> <NavigationProperty Name="feederExceptions" FromRole="DpfSolution" ToRole="feederExceptions" Relationship="DmsDataModel.DpfSolution_feederExceptions"/> <NavigationProperty Name="voltageViolations" FromRole="DpfSolution" ToRole="voltageViolations" Relationship="DmsDataModel.DpfSolution_voltageViolations"/> <NavigationProperty Name="overloadViolations" FromRole="DpfSolution" ToRole="overloadViolations" Relationship="DmsDataModel.DpfSolution_overloadViolations"/> <NavigationProperty Name="voltageOutputs" FromRole="DpfSolution" ToRole="voltageOutputs" Relationship="DmsDataModel.DpfSolution_voltageOutputs"/> <NavigationProperty Name="MeasurementIslands" FromRole="DpfSolution" ToRole="MeasurementIslands" Relationship="DmsDataModel.DpfSolution_MeasurementIslands"/> <NavigationProperty Name="FeederLosses" FromRole="DpfSolution" ToRole="FeederLosses" Relationship="DmsDataModel.DpfSolution_FeederLosses"/> <NavigationProperty Name="TfLosses" FromRole="DpfSolution" ToRole="TfLosses" Relationship="DmsDataModel.DpfSolution_TfLosses"/> </EntityType> </pre>
Station Exception	<pre> <EntityType Name="StationException"> <Key><PropertyRef Name="StationName"/></Key> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="AreaName" Type="Edm.String"/> <Property Name="NumberOfEnergizedCustomers" Nullable="false" Type="Edm.Int32"/> <Property Name="NumberOfAbnormalDevices" Nullable="false" Type="Edm.Int32"/> <Property Name="HighestVoltageViolationPercent" Type="Edm.Double"/> <Property Name="LowestVoltageViolationPercent" Type="Edm.Double"/> <Property Name="LargestVoltageImbalancePercent" Type="Edm.Double"/> <Property Name="LargestFlowImbalancePercent" Type="Edm.Double"/> <Property Name="OverloadLevel" Type="Edm.Int16"/> <Property Name="LoadingPercent" Type="Edm.Double"/> <Property Name="GroundReturnCurrentPercent" Type="Edm.Double"/> <Property Name="DpfStatus" Type="Edm.String"/> </EntityType> </pre>

Name	Entity Type
Feeder Exception	<pre> -<EntityType Name="FeederException"> -<Key><PropertyRef Name="FeederName"/> <PropertyRef Name="StationName"/></Key> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="FeederId" Type="Edm.String"/> <Property Name="FeederName" Nullable="false" Type="Edm.String"/> <Property Name="AreaName" Type="Edm.String"/> <Property Name="NumberOfDeenergizedCustomers" Type="Edm.Int32"/> <Property Name="NumberOfAbnormalDevices" Type="Edm.Int32"/> <Property Name="HighestVoltageViolationPercent" Type="Edm.Double"/> <Property Name="LowestVoltageViolationPercent" Type="Edm.Double"/> <Property Name="LargestVoltageImbalancePercent" Type="Edm.Double"/> <Property Name="LargestFlowImbalancePercent" Type="Edm.Double"/> <Property Name="OverLoadLevel" Type="Edm.Int16"/> <Property Name="LoadingPercent" Type="Edm.Double"/> <Property Name="GroundReturnCurrentPercent" Type="Edm.Double"/> <Property Name="DpfStatus" Type="Edm.String"/> </EntityType> </pre>
Voltage Violation	<pre> -<EntityType Name="VoltageViolation"> -<Key><PropertyRef Name="Id"/> <PropertyRef Name="StationName"/></Key> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="LimitValue" Type="Edm.String"/> <Property Name="VoltageValue" Type="Edm.String"/> </EntityType> </pre>
Overload Violation	<pre> -<EntityType Name="OverloadViolation"> -<Key><PropertyRef Name="Id"/> <PropertyRef Name="StationName"/></Key> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="LimitValue" Type="Edm.String"/> <Property Name="FlowValue" Type="Edm.String"/> </EntityType> </pre>
VoltageOutput	<pre> -<EntityType Name="VoltageOutput"> -<Key><PropertyRef Name="Id"/> <PropertyRef Name="StationName"/></Key> <Property Name="StationName" Nullable="false" Type="Edm.String"/> <Property Name="Id" Nullable="false" Type="Edm.String"/> <Property Name="VoltA" Type="Edm.String"/> <Property Name="VoltB" Type="Edm.String"/> <Property Name="VoltC" Type="Edm.String"/> </EntityType> </pre>

Name	Entity Type
DpfMeasurementIsland	<pre> <EntityType Name="DpfMeasurementIsland"> <Key><PropertyRef Name="MeasurementIslandIndex"/> <PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="MeasurementIslandName" Type="Edm.String"/> <Property Name="MeasurementIslandIndex" Nullable="false" Type="Edm.String"/> <Property Name="MeasurementIslandType" Type="Edm.String"/> <Property Name="IsShowable" Type="Edm.Boolean"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="FeederName" Type="Edm.String"/> <Property Name="BranchName" Type="Edm.String"/> <Property Name="LoadAllocationType" Type="Edm.String"/> <Property Name="RealPowerFlowDirectionPhaseA" Type="Edm.String"/> <Property Name="RealPowerFlowDirectionPhaseB" Type="Edm.String"/> <Property Name="RealPowerFlowDirectionPhaseC" Type="Edm.String"/> <Property Name="ReactivePowerFlowDirectionPhaseA" Type="Edm.String"/> <Property Name="ReactivePowerFlowDirectionPhaseB" Type="Edm.String"/> <Property Name="ReactivePowerFlowDirectionPhaseC" Type="Edm.String"/> <Property Name="ApparentPowerFlowDirectionPhaseA" Type="Edm.String"/> <Property Name="ApparentPowerFlowDirectionPhaseB" Type="Edm.String"/> <Property Name="ApparentPowerFlowDirectionPhaseC" Type="Edm.String"/> <Property Name="RealPowerConvergence" Type="Edm.String"/> <Property Name="ReactivePowerConvergence" Type="Edm.String"/> <Property Name="ApparentPowerConvergence" Type="Edm.String"/> <Property Name="TotalRealPower" Type="Edm.Double"/> <Property Name="TotalReactivePower" Type="Edm.Double"/> <Property Name="Mismatch" Type="Edm.Double"/> <Property Name="NumberOfLoadsPhaseA" Type="Edm.Int32"/> <Property Name="NumberOfLoadsPhaseB" Type="Edm.Int32"/> <Property Name="NumberOfLoadsPhaseC" Type="Edm.Int32"/> <Property Name="NumberOfLoadsBalanced3Phase" Type="Edm.Int32"/> <Property Name="NumberOfLoadsNon3Phase" Type="Edm.Int32"/> <Property Name="NumberOfLoadsOpenWyeDelta" Type="Edm.Int32"/> </EntityType> </pre>
DpfFeederLosses	<pre> <EntityType Name="DpfFeederLosses"> <Key><PropertyRef Name="FeederId"/> <PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="FeederId" Nullable="false" Type="Edm.String"/> <Property Name="FeederName" Type="Edm.String"/> <Property Name="RealPowerLoss" Type="Edm.String"/> <Property Name="ReactivePowerLoss" Type="Edm.String"/> </EntityType> </pre>
DpfTfLosses	<pre> <EntityType Name="DpfTfLosses"> <Key><PropertyRef Name="PowerTransformerId"/> <PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="PowerTransformerId" Nullable="false" Type="Edm.String"/> <Property Name="PowerTransformerName" Type="Edm.String"/> <Property Name="RealPowerLoss" Type="Edm.String"/> <Property Name="ReactivePowerLoss" Type="Edm.String"/> </EntityType> </pre>

Name	Entity Type
DsoFeederParameters	<pre> -<EntityType Name="DsoFeederParameters"> -<Key><PropertyRef Name="FeederId"/> <PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="FeederId" Nullable="false" Type="Edm.String"/> <Property Name="FeederName" Type="Edm.String"/> <Property Name="IsAdvancedAppEnabled" Type="Edm.Boolean"/> <Property Name="ProblemFormulation" Type="Edm.String"/> </EntityType> </pre>
DsoTfParameters	<pre> -<EntityType Name="DsoTfParameters"> -<Key><PropertyRef Name="PowerTransformerId"/> <PropertyRef Name="SolutionId"/></Key> <Property Name="SolutionId" Nullable="false" Type="Edm.String"/> <Property Name="StationId" Type="Edm.String"/> <Property Name="StationName" Type="Edm.String"/> <Property Name="PowerTransformerId" Nullable="false" Type="Edm.String"/> <Property Name="PowerTransformerName" Type="Edm.String"/> <Property Name="IsAdvancedAppEnabled" Type="Edm.Boolean"/> <Property Name="ProblemFormulation" Type="Edm.String"/> </EntityType> </pre>

3. DMS Query Service

The DMS Query Service allows external applications to send ad hoc DMS queries to the system. The query interface supports both GET and POST calls.

3.1. GET Call

Important

The DMS Query Service GET call is obsolete. This call will be removed in a future release for security concerns. Use the POST call instead.

For a GET query, the return is in XML format.

The form of the call is:

```
http://localhost:8801/dms/data/query/DmsQuery?query="query string"
```

The query string has the following format (the WHERE clause is optional):

```
SELECT xxx FROM xxx WHERE xxx
```

For example:

```
http://localhost:8801/dms/data/query/DmsQuery?query='SELECT%20ID,NAME,AREANAME,NUMDEENG,NUMABTOT%20FROM%20RT.SUB%20WHERE%20NAME=RAVEN'
```

This returns:

```
<EntityList>
  <Entity>
    <ID>8163417</ID>
    <NAME>RAVEN</NAME>
    <AREANAME>WST</AREANAME>
    <NUMDEENG>7006</NUMDEENG>
    <NUMABTOT>8</NUMABTOT>
  </Entity>
</EntityList>
```

3.2. POST Call

For a POST query, the payload can be in either XML or JSON format depending on the content type that is specified.

The form of the call is:

```
http://localhost:8801/dms/data/query/DmsQuery
```

The POST body can be in either XML or JSON format.

This example uses XML format:

```
Content-Type: application/xml

<?xml version="1.0" encoding="utf-8"?>
<QueryRequest xmlns="http://com.alstom.dms.rest">
  <Select>ID,NAME,AREANAME,NUMDEENG,NUMABTOT</Select>
  <From>RT.SUB</From>
  <Where>NAME=RAVEN</Where>
</QueryRequest>
```

This returns:

```
<EntityList>
  <Entity>
    <ID>8163417</ID>
    <NAME>RAVEN</NAME>
    <AREANAME>WST</AREANAME>
    <NUMDEENG>7006</NUMDEENG>
    <NUMABTOT>8</NUMABTOT>
  </Entity>
</EntityList>
```

This example uses JSON format:

```
Content-Type: application/json
{
  "Select": "ID,NAME,AREANAME,NUMDEENG,NUMABTOT",
  "From": "RT.SUB",
  "Where": "NAME=RAVEN"
}
```

This returns:

```
[
  {
    "ID": "8163417",
    "NAME": "RAVEN",
    "AREANAME": "WST",
    "NUMDEENG": "7006",
    "NUMABTOT": "8"
  }
]
```


4. DMS Control Service

The DMS Control service allows external applications to update DMS data and request services from the DMS server. The service is implemented as a set of REST POST calls. Service calls support both XML and JSON formats and can be accessed via <http://localhost:8801/dms/services/>.

The following calls are currently supported:

- **BlockPrediction:** Blocks outage prediction for a device.
- **ConfirmDeviceState:** Confirms a device status change and then updates the device state for a distribution transformer, cut, switching device, or capacitor. Provide the device station, type, ID, status, and phases to change.
- **DerAnalog:** Updates analog measurement values for a distributed energy resource (DER).
- **DerAnalogList:** Updates analog measurement values for a number of DERs.
- **DerControlStatus:** Updates the control status for a DER.
- **DerControlStatusList:** Updates control statuses for a number of DERs.
- **DerLimit:** Updates limits for a DER.
- **DerLimitList:** Updates limits for a number of DERs.
- **DerStatus:** Updates the alarm, connection, and communication statuses for a DER.
- **DerStatusList:** Updates the alarm, connection, and communication statuses for a number of DERs.
- **DerStatusSim:** Updates the alarm, connection, or communication status for a DER in the simulator mode.
- **DeviceData:** Updates device data in the real-time mode.
- **DeviceDataSim:** Updates device data in the simulator mode.
- **DnafRequest:** Sends a request to run a DNAF application.
- **DnafStudyRequest:** Sends a request to run a DNAF application in study mode.
- **DpfRequest:** Sends a request to run a DPF application.
- **DpfStudyRequest:** Sends a DPF request to the Study server.
- **GetStudySavecases:** Gets a list of study server savecases that exist in a folder.
- **GroupDeviceData:** Sends a request to update data for one or more devices.
- **ImplementDnafPluginDisplay:** Sends an LVM System Monitor configuration implementation request to the DNAF Display plugin. This call is used for internal purposes.
- **InitializeSimToNominal:** Resets the simulator mode to the nominal system state.
- **InitializeStudyServer:** Initializes the study mode.
- **ManagedDataEntryRequest:** Updates DMS or OMS data.
- **MoveMobileSubStation:** Moves a mobile substation.
-

PlaceCut: Places a cut in the real-time mode.

- **PlaceCutStudy:** Places a cut in the study mode.
- **PlaceFaultRT:** Places a fault in the real-time mode.
- **PlaceFaultSIM:** Places a fault in the simulator mode.
- **PlaceJumper:** Places a jumper in the real-time mode.
- **PlaceMobileSubStation:** Places a mobile substation.
- **PlaceRemoveTagST:** Creates, modifies, or deletes a tag in study mode.
- **PlaceSwitch:** Places a temporary switch in the real-time mode.
- **PlaceTemporaryGround:** Places a temporary ground in the real-time mode.
- **RemoveCut:** Removes a cut in the real-time mode.
- **RemoveJumper:** Removes a jumper in the real-time mode.
- **RemoveMobileSubStation:** Removes a mobile substation.
- **RemoveTemporaryEventRT:** Removes a fault or temporary ground in the real-time mode.
- **RemoveTemporaryEventSIM:** Removes a fault or temporary ground in the simulator mode.
- **RequestResolveQuery:** Sends a DMS query.
- **ResetStudyServer:** Cleans up the dynamics for the Study server.
- **ResolveQuery:** Sends a query to the DNAF server for DMS data. If the DNAF server is not available, the query is routed to the DMS server.
- **SaveStudyCase:** Creates a Study server savecase.
- **SendDataEntry:** Updates DNAF data in the real-time mode. This call is used for internal purposes.
- **SendDataEntrySim:** Updates DNAF data in the simulator mode. This call is used for internal purposes.
- **SendDnafPluginDisplayRequest:** Sends an LVM System Monitor configuration request to the DNAF Display plugin. This call is used for internal purposes.
- **SetDeviceState:** Changes device status for a distribution transformer, cut, switching device, or capacitor. Provide the device station, type, ID, status, and phases to change.
- **StudySavecaseRestore:** Retrieves the specified study savecase.
- **SwitchStatus:** Updates the switch status (open/closed).
- **TapPosition:** Updates the tap position for a tap changing transformer.
- **TempSwitchStatus:** Updates the status (open/closed) for temporary switch (jumper).
- **UpdateSwitchStatusInSim:** Updates the switching device status in the simulator mode.

The following web services are based on the DMS Control service

- **Annotation Web Service:** Creates, updates, and deletes annotations.
- **Switching Order Web Service:** Updates switching orders.

- **Schedules Web Service:** Sends and updates schedules.

In addition, an event notification mechanism on the DMS interface server can send notifications about interested events to external applications. The event is sent via a REST call to the registered URLs.

The following events are currently supported:

- **SwitchStatus:** Notifications about the switch status changes.
- **DnafStatus:** Notifications about the DNAF DPF results.

4.1. Authorization Header

To provide additional security for control service calls, an "authorization" header is used to authenticate REST calls from external applications for all POST update requests.

The authorization header format is as follows:

```
"authorization: DmsISClient:72bPJGtWk0Fuv6lKndo/KA=="
```

The authorization header includes the client ID and the hashed key, separated by a colon (:).

Note

The hashed key must be generated using the relative request path and client key using the HmacSHA256 algorithm, and the result must be Base64 encoded.

The client ID is configured through the DmsInterfaceServerConfig.xml file (see [DMS Interface Server Configuration](#)) and the hashed key is calculated by the request URI and client key (also in the same configuration file). The URI used for token generation must contain only the request path (relative URI) and must not include protocol, hostname, or port.

Example of valid URI:

```
/dms/services/schedules/set
```

Invalid URI (do not use):

```
https://hostname:port/dms/services/schedules/set
```

The following is sample Java code for setting the key in the request header:

```
import javax.crypto.Mac;
import javax.crypto.spec.SecretKeySpec;
public static String USER_ID = "DmsISClient";
public static String USER_KEY = "Test123";

public static String getAuthToken(String uri) throws Exception
```

```

{
    SecretKeySpec keySpec = new SecretKeySpec(USER_KEY.getBytes("ASCII"), "HmacSHA256");
    Mac = Mac.getInstance("HmacSHA256");
    mac.init(keySpec);
    byte[] result = mac.doFinal(uri.getBytes("ASCII"));
    BASE64Encoder encoder = new BASE64Encoder();
    return encoder.encode(result);
}
...
HttpHeaders headers = new HttpHeaders();
String authString = RestClientUtil.USER_ID + ":" + getAuthToken(requestUrl);
headers.set("Authorization", authString);

```

4.2. Service API

The following tables list the details of the service calls. For more details, refer to <http://localhost:8801/dms/services/help>.

The fields in **bold** are required.

Note

The sample payloads below are in XML format; JSON format is also supported. As long as the correct content type ("application/xml" versus "application/json") is specified, the server will parse the request correctly.

4.2.1. BlockPrediction Service Call

POST

/dms/services/BlockPrediction

Blocks outage prediction for a device.

Note

The DeviceId must exist in the DMS server.

Sample Payload	Return
<pre> <BlockPredictionContext xmlns="http://com.alstom.dms.rest"> <StationName>Raven</StationName> <DeviceId>1234</DeviceId> <DeviceType>Fuse</DeviceType> <Phases>ABC</Phases> </BlockPredictionContext> </pre>	None

4.2.2. ConfirmDeviceState Service Call

POST

/dms/services/ConfirmDeviceState

Used to confirm the device status change and then update the device state for a distribution transformer, cut, switching device, or capacitor.

Note
The DeviceId must exist in the DMS server.

Note
The DeviceType value must follow the associated DeviceType coded translation item key.

Sample Payload	Return
<pre><DeviceState xmlns="http://com.alstom.dms.rest"> <StationName>String content</StationName> <DeviceId>String content</DeviceId> <Status>true</Status> <DeviceType>2147483647</DeviceType> <Phases>String content</Phases> <Properties> <KeyValueOfstringstring xmlns="http://schemas.microsoft.com/2003/10/Serialization/Array s"> <Key>String content</Key> <Value>String content</Value> </KeyValueOfstringstring> <KeyValueOfstringstring xmlns="http://schemas.microsoft.com/2003/10/Serialization/Array s"> <Key>String content</Key> <Value>String content</Value> </KeyValueOfstringstring> </Properties> <isConfirmDeviceState>true</isConfirmDeviceState> <OperationTime>1999-05-31T11:20:00</OperationTime> </DeviceState></pre>	<pre><ConfirmResponse xmlns="http://com.alstom.dms.rest"> <Status>true</Status> <ErrorMessage>String content</ErrorMessage> </ConfirmResponse></pre>

4.2.3. DerAnalog Service Call

POST

/dms/services/DerAnalog

Updates analog measurement values for a DER.

Sample Payload	Return
<pre> <DerAnalogUpdate xmlns="http://com.alstom.dms.rest"> <Id>String content</Id> <PhaseAkV>1.26743237E+15</PhaseAkV> <PhaseAkVar>1.26743237E+15</PhaseAkVar> <PhaseAkW>1.26743237E+15</PhaseAkW> <PhaseBkV>1.26743237E+15</PhaseBkV> <PhaseBkVar>1.26743237E+15</PhaseBkVar> <PhaseBkW>1.26743237E+15</PhaseBkW> <PhaseCkV>1.26743237E+15</PhaseCkV> <PhaseCkVar>1.26743237E+15</PhaseCkVar> <PhaseCkW>1.26743237E+15</PhaseCkW> <QualityFlags>2147483647</QualityFlags> <StateOfCharge>1.26743237E+15</StateOfCharge> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerAnalogUpdate> </pre>	None

4.2.4. DerAnalogList Service Call

POST

```
/dms/services/DerAnalogList
```

Updates analog measurement values for a number of DERs.

Sample Payload	Return
<pre> <ArrayOfDerAnalogUpdate xmlns="http://com.alstom.dms.rest"> <DerAnalogUpdate> <Id>String content</Id> <PhaseAkV>1.26743237E+15</PhaseAkV> <PhaseAkVar>1.26743237E+15</PhaseAkVar> <PhaseAkW>1.26743237E+15</PhaseAkW> <PhaseBkV>1.26743237E+15</PhaseBkV> <PhaseBkVar>1.26743237E+15</PhaseBkVar> <PhaseBkW>1.26743237E+15</PhaseBkW> <PhaseCkV>1.26743237E+15</PhaseCkV> <PhaseCkVar>1.26743237E+15</PhaseCkVar> <PhaseCkW>1.26743237E+15</PhaseCkW> <QualityFlags>2147483647</QualityFlags> <StateOfCharge>1.26743237E+15</StateOfCharge> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerAnalogUpdate> <DerAnalogUpdate> <Id>String content</Id> <PhaseAkV>1.26743237E+15</PhaseAkV> <PhaseAkVar>1.26743237E+15</PhaseAkVar> <PhaseAkW>1.26743237E+15</PhaseAkW> <PhaseBkV>1.26743237E+15</PhaseBkV> <PhaseBkVar>1.26743237E+15</PhaseBkVar> <PhaseBkW>1.26743237E+15</PhaseBkW> <PhaseCkV>1.26743237E+15</PhaseCkV> <PhaseCkVar>1.26743237E+15</PhaseCkVar> <PhaseCkW>1.26743237E+15</PhaseCkW> <QualityFlags>2147483647</QualityFlags> <StateOfCharge>1.26743237E+15</StateOfCharge> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerAnalogUpdate> </ArrayOfDerAnalogUpdate> </pre>	None

4.2.5. DerControlStatus Service Call

POST

/dms/services/DerControlStatus

Updates the control status for a DER.

Sample Payload	Return
<pre> <DerControlResponse xmlns="http://com.alstom.dms.rest"> <Control>String content</Control> <EnergyResourceId>String content</EnergyResourceId> <PlanId>String content</PlanId> <StationName>String content</StationName> <Status>String content</Status> <Step>2147483647</Step> <TimeAcknowledged>9223372036854775807</TimeAcknowledged> </DerControlResponse> </pre>	None

4.2.6. DerControlStatusList Service Call

POST

```
/dms/services/DerControlStatusList
```

Updates control statuses for a number of DERs.

Sample Payload	Return
<pre><ArrayOfDerControlResponse xmlns="http://com.alstom.dms.rest"> <DerControlResponse> <Control>String content</Control> <EnergyResourceId>String content</EnergyResourceId> <PlanId>String content</PlanId> <StationName>String content</StationName> <Status>String content</Status> <Step>2147483647</Step> <TimeAcknowledged>9223372036854775807</TimeAcknowledged> </DerControlResponse> <DerControlResponse> <Control>String content</Control> <EnergyResourceId>String content</EnergyResourceId> <PlanId>String content</PlanId> <StationName>String content</StationName> <Status>String content</Status> <Step>2147483647</Step> <TimeAcknowledged>9223372036854775807</TimeAcknowledged> </DerControlResponse> </ArrayOfDerControlResponse></pre>	None

4.2.7. DerLimit Service Call

POST

```
/dms/services/DerLimit
```

Updates limits for a DER.

Sample Payload	Return
<pre><DerLimitUpdate xmlns="http://com.alstom.dms.rest"> <Id>String content</Id> <MaxChargeRatekW>1.26743237E+15</MaxChargeRatekW> <MaxDischargeRatekW>1.26743237E+15</MaxDischargeRatekW> <MaxkVA>1.26743237E+15</MaxkVA> <MaxkVar>1.26743237E+15</MaxkVar> <MaxkVarNeg>1.26743237E+15</MaxkVarNeg> <MaxkW>1.26743237E+15</MaxkW> <MinPfOverExcited>1.26743237E+15</MinPfOverExcited> <MinPfUnderExcited>1.26743237E+15</MinPfUnderExcited> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerLimitUpdate></pre>	None

4.2.8. DerLimitList Service Call

POST

```
/dms/services/DerLimitList
```

Updates limits for a number of DERs.

Sample Payload	Return
<pre><ArrayOfDerLimitUpdate xmlns="http://com.alstom.dms.rest"> <DerLimitUpdate> <Id>String content</Id> <MaxChargeRatekW>1.26743237E+15</MaxChargeRatekW> <MaxDischargeRatekW>1.26743237E+15</MaxDischargeRatekW> <MaxkVA>1.26743237E+15</MaxkVA> <MaxkVar>1.26743237E+15</MaxkVar> <MaxkVarNeg>1.26743237E+15</MaxkVarNeg> <MaxkW>1.26743237E+15</MaxkW> <MinPfOverExcited>1.26743237E+15</MinPfOverExcited> <MinPfUnderExcited>1.26743237E+15</MinPfUnderExcited> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerLimitUpdate> <DerLimitUpdate> <Id>String content</Id> <MaxChargeRatekW>1.26743237E+15</MaxChargeRatekW> <MaxDischargeRatekW>1.26743237E+15</MaxDischargeRatekW> <MaxkVA>1.26743237E+15</MaxkVA> <MaxkVar>1.26743237E+15</MaxkVar> <MaxkVarNeg>1.26743237E+15</MaxkVarNeg> <MaxkW>1.26743237E+15</MaxkW> <MinPfOverExcited>1.26743237E+15</MinPfOverExcited> <MinPfUnderExcited>1.26743237E+15</MinPfUnderExcited> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerLimitUpdate> </ArrayOfDerLimitUpdate></pre>	None

4.2.9. DerStatus Service Call

POST

```
/dms/services/DerStatus
```

Updates the alarm, communication, and connection statuses for a DER.

Sample Payload	Return
<pre><DerStatusUpdate xmlns="http://com.alstom.dms.rest"> <AlarmStatus>2147483647</AlarmStatus> <CommunicationStatus>2147483647</CommunicationStatus> <ConnectionStatus>2147483647</ConnectionStatus> <Id>String content</Id> <StateOfCharge>1.26743237E+15</StateOfCharge> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerStatusUpdate></pre>	None

4.2.10. DerStatusList Service Call

POST

```
/dms/services/DerStatusList
```

Updates the alarm, communication, and connection statuses for a number DERs.

Sample Payload	Return
<pre><ArrayOfDerStatusUpdate xmlns="http://com.alstom.dms.rest"> <DerStatusUpdate> <AlarmStatus>2147483647</AlarmStatus> <CommunicationStatus>2147483647</CommunicationStatus> <ConnectionStatus>2147483647</ConnectionStatus> <Id>String content</Id> <StateOfCharge>1.26743237E+15</StateOfCharge> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerStatusUpdate> <DerStatusUpdate> <AlarmStatus>2147483647</AlarmStatus> <CommunicationStatus>2147483647</CommunicationStatus> <ConnectionStatus>2147483647</ConnectionStatus> <Id>String content</Id> <StateOfCharge>1.26743237E+15</StateOfCharge> <StationName>String content</StationName> <UpdateTime>9223372036854775807</UpdateTime> </DerStatusUpdate> </ArrayOfDerStatusUpdate></pre>	None

4.2.11. DerStatusSim Service Call

POST

```
/dms/services/DerStatusSim
```

Updates the alarm, communication, or connection status for a DER.

Sample Payload	Return
<pre>{ "StationName" : "RAVEN", "DeviceId" : "RAVEN_ERS1", "Attribute" : 1, //1 - ConnectionStatus, 2 - CommunicationStatus, 3 - AlarmStatus "Status" : 0 }</pre>	<pre>{ "Status":true, "Message":"String content" }</pre>

4.2.12. DeviceData Service Call

POST

/dms/services/DeviceData

Used to update any device data. You must provide the device ID, type, and the list of fields to change.

Note

The DeviceId and FieldName must exist in the DMS server.

The OldValue is optional and can be omitted if there is none.

Sample Payload	Return
<pre><?xml version="1.0" encoding="UTF-8" standalone="yes"?> <DeviceData xmlns="http://com.alstom.dms.rest"> <StationName>RAVEN</StationName> <DeviceId>925A3D8E-0FD1-4617-ADAB-EC2CAF58A466</DeviceId> <DeviceType>TF</DeviceType> <FieldList> <DeviceFieldEntry> <FieldName>TAPEND</FieldName> <NewValue>1</NewValue> <OldValue></OldValue> </DeviceFieldEntry> ... </FieldList> </DeviceData></pre>	None

4.2.13. DeviceDataSim Service Call

POST

/dms/services/DeviceDataSim

Used to update any device data in the simulator mode. You must provide the device ID, type, and the list of fields to change.

###

Note

The DeviceId and FieldName must exist in the DMS server.

The OldValue is optional and can be omitted if there is none.

Sample Payload	Return
The DeviceDataSim call uses the DeviceData schema. For the request example, see DeviceData Service Call .	None

4.2.14. DnafRequest Service Call

POST

/dms/services/DnafRequest

Request to run a DNAF application. The Options group applies to the DnafStudyRequest call only.

Sample Payload	Return
<pre><DnafRequestData xmlns="http://com.alstom.dms.rest"> <StationName>String content</StationName> <ComponentId>String content</ComponentId> <ComponentType>2147483647</ComponentType> <DnafFunction>2147483647</DnafFunction> <ProblemFormulation>String content</ProblemFormulation> <DateTime>1999-05-31T11:20:00</DateTime> <LoadAllocate>2147483647</LoadAllocate> <ControlSetting>String content</ControlSetting> <Options> <AllDevices>32767</AllDevices> <Battery>32767</Battery> <Biomass>32767</Biomass> <CapacitorControl>Local</CapacitorControl> <Cogeneration>32767</Cogeneration> <FuelCell>32767</FuelCell> <Gas>32767</Gas> <GenDerOutput>UseExisting</GenDerOutput> <Hydro>32767</Hydro> <MicroTurbine>32767</MicroTurbine> <Other>32767</Other> <Solar>32767</Solar> <TapControl>Local</TapControl> <Wind>32767</Wind> </Options> </DnafRequestData></pre>	<pre><DnafRequestResponse xmlns="http://com.alstom.dms.rest"> <GUID>String content</GUID> </DnafRequestResponse></pre> Guid is returned from the DMS Server.

4.2.15. DnafStudyRequest Service Call

POST

/dms/services/DnafStudyRequest

Request to run a DNAF application in study mode. The Options group includes configurations from

the Study Mode Options dialog on the study mode DNAF Solution display. If the Options group is provided, the ControlSetting field is ignored.

Sample Payload	Return
<pre><DnafRequestData xmlns="http://com.alstom.dms.rest"> <StationName>String content</StationName> <ComponentId>String content</ComponentId> <ComponentType>2147483647</ComponentType> <DnafFunction>2147483647</DnafFunction> <ProblemFormulation>String content</ProblemFormulation> <DateTime>1999-05-31T11:20:00</DateTime> <LoadAllocate>2147483647</LoadAllocate> <ControlSetting>String content</ControlSetting> <Options> <AllDevices>32767</AllDevices> <Battery>32767</Battery> <Biomass>32767</Biomass> <CapacitorControl>Local</CapacitorControl> <Cogeneration>32767</Cogeneration> <FuelCell>32767</FuelCell> <Gas>32767</Gas> <GenDerOutput>UseExisting</GenDerOutput> <Hydro>32767</Hydro> <MicroTurbine>32767</MicroTurbine> <Other>32767</Other> <Solar>32767</Solar> <TapControl>Local</TapControl> <Wind>32767</Wind> </Options> </DnafRequestData></pre>	<pre><DnafRequestResponse xmlns="http://com.alstom.dms.rest"> <GUID>String content</GUID> </DnafRequestResponse></pre> Guid is returned from the DMS Server.

4.2.16. DpfRequest Service Call

POST

/dms/services/DpfRequest

Request to run a DPF application.

Sample Payload	Return
<pre><DpfRequestData xmlns="http://com.alstom.dms.rest"> <DateTime>1999-05-31T11:20:00</DateTime> <LoadAllocate>2147483647</LoadAllocate> <StationName>String content</StationName> </DpfRequestData></pre>	<pre><DpfRequestResponse xmlns="http://com.alstom.dms.rest"> <GUID>String content</GUID> </DpfRequestResponse></pre> Guid returned from the DMS Server.

4.2.17. DpfStudyRequest Service Call

POST

/dms/services/DpfStudyRequest

Request to run a DPF application in Study mode.

LoadAllocate is always 1.

ControllList and ViolationList are optional.

Sample Payload	Return
<pre><DpfStudyRequestData xmlns="http://com.alstom.dms.rest"> <ControllList> <OptimizationControl> <ControlMRID>String content</ControlMRID> <ConsumerMRID>String content</ConsumerMRID> <ControlPhase>String content</ControlPhase> <ActivePowerDelta>1.26743233E+15</ActivePowerDelta> <ReactivePowerDelta>1.26743233E+15</ReactivePowerDelta> <DropSpecified>true</DropSpecified> </OptimizationControl> <OptimizationControl> <ControlMRID>String content</ControlMRID> <ConsumerMRID>String content</ConsumerMRID> <ControlPhase>String content</ControlPhase> <ActivePowerDelta>1.26743233E+15</ActivePowerDelta> <ReactivePowerDelta>1.26743233E+15</ReactivePowerDelta> <DropSpecified>true</DropSpecified> </OptimizationControl> </ControllList> <DateTime>1999-05-31T11:20:00</DateTime> <LoadAllocate>2147483647</LoadAllocate> <StationName>String content</StationName> <ViolationList> <VoltageViolation> <EquipmentMRID>String content</EquipmentMRID> <ViolationMRID>String content</ViolationMRID> <Phase>String content</Phase> </VoltageViolation> <VoltageViolation> <EquipmentMRID>String content</EquipmentMRID> <ViolationMRID>String content</ViolationMRID> <Phase>String content</Phase> </VoltageViolation> </ViolationList> </DpfStudyRequestData></pre>	<pre><DpfRequestResponse xmlns="http://com.alstom.dms.rest"> <GUID>String content</GUID> </DpfRequestResponse></pre> Guid returned from the DMS Server.

4.2.18. GetStudySavecases Service Call

POST

```
/dms/services/GetStudySavecases
```

Gets a list of study server save cases that exist in a folder.

Sample Payload	Return
<pre><GetStudySavecasesData xmlns="http://com.alstom.dms.rest"> <Path>String content</Path> </GetStudySavecasesData></pre>	<pre><ArrayOfstring xmlns="http://schemas.microsoft.com/2003/10/ Serialization/Arrays"> <string>String content</string> <string>String content</string> </ArrayOfstring></pre>

4.2.19. GroupDeviceDataService Call

POST

/dms/services/GroupDeviceData

Sends a request to the DNAF server to update data for one or more devices.

Sample Payload	Return
<pre><GroupDeviceData xmlns="http://com.alstom.dms.rest"> <DeviceDataList> <DeviceData> <StationName>String content</StationName> <DeviceId>String content</DeviceId> <DeviceType>String content</DeviceType> <FieldList> <DeviceFieldEntry> <FieldName>String content</FieldName> <NewValue>String content</NewValue> <OldValue>String content</OldValue> </DeviceFieldEntry> <DeviceFieldEntry> <FieldName>String content</FieldName> <NewValue>String content</NewValue> <OldValue>String content</OldValue> </DeviceFieldEntry> </FieldList> </DeviceData> <DeviceData> <StationName>String content</StationName> <DeviceId>String content</DeviceId> <DeviceType>String content</DeviceType> <FieldList> <DeviceFieldEntry> <FieldName>String content</FieldName> <NewValue>String content</NewValue> <OldValue>String content</OldValue> </DeviceFieldEntry> <DeviceFieldEntry> <FieldName>String content</FieldName> <NewValue>String content</NewValue> <OldValue>String content</OldValue> </DeviceFieldEntry> </FieldList> </DeviceData> </DeviceDataList> </GroupDeviceData></pre>	The response body is empty.

4.2.20. ImplementDnafPluginDisplayService Call

POST

/dms/services/ImplementDnafPluginDisplay

Sends an LVM System Monitor configuration implementation request to the DNAF Display plugin. This call is used for internal purposes.

Sample Payload	Return
<pre><ImplementDnafPluginDisplayRequest xmlns="http://com.alstom.dms.rest"> <Command>String content</Command> <Option>String content</Option> <Comments>String content</Comments> <Minutesoldplan>2147483647</Minutesoldplan> </ImplementDnafPluginDisplayRequest></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.21. InitializeSimToNominal Service Call

POST

/dms/services/InitializeSimToNominal

Resets the simulator mode to the nominal system state. This also results it resetting the real-time mode to the nominal system state.

Sample Payload	Return
N/A	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.22. InitializeStudyServer Service Call

POST

/dms/services/InitializeStudyServer

Initializes the study mode to the nominal or real-time system state for the specified substation. You can specify whether to apply mobile substation and dynamic schedule data from the real-time server.

Note

To initialize the study mode, the Study server and the DMS Interface server must be hosted on

the same machine.

Sample Payload	Return
<pre><StudyServerInitializationData xmlns="http://com.alstom.dms.rest"> <CopyMobileSubStation>true</CopyMobileSubStation> <DynSchedule>String content</DynSchedule> <IsNominalState>true</IsNominalState> <StationName>String content</StationName> </StudyServerInitializationData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.23. ManagedDataEntryRequest Service Call

POST

/dms/services/ManagedDataEntryRequest

Updates DMS or OMS data.

Sample Payload	Return
<pre>{ "ObjectNameKey": "TEMPFI1518", "SenderName": "CREW", "UserName": "DMSALL", "Historical": false, "FieldList": [{ "FieldName": "PRIMARYEMPLOYEE", "NewValue": "FI1518" } , { "FieldName": "PRIMARYEMPLOYEE", "NewValue": "N Scott Travis" } , { "FieldName": "RADIONUMBER", "NewValue": "" }] }</pre>	The Response body is empty.

4.2.24. MoveMobileSubStation Service Call

POST

/dms/services/MoveMobileSubStation

Moves a mobile substation.

Sample Payload	Return
<pre> <MobileSubStationEventData xmlns="http://com.ge.dms.rest"> <GeoName>String content</GeoName> <NearestStation>String content</NearestStation> <StationName>String content</StationName> <X>1.26743233E+15</X> <Y>1.26743233E+15</Y> </MobileSubStationEventData> Sample JSON payload: { "StationName": "MOBILE2", "NearestStation": "RAVEN", "X": "935703", "Y": "812673", "GeoName": "DETAIL" }</pre>	<pre> <Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.25. PlaceCut Service Call

POST

```
/dms/services/PlaceCut
```

Places a cut in the real-time mode.

Sample Payload	Return
<pre> <PlaceCutEventData xmlns="http://com.alstom.dms.rest"> <Comment>String content</Comment> <CutID>String content</CutID> <CutLength>1.26743233E+15</CutLength> <CutName>String content</CutName> <GeoName>String content</GeoName> <IsClosed>true</IsClosed> <LineID>String content</LineID> <Phases>String content</Phases> <StationName>String content</StationName> </PlaceCutEventData></pre>	<pre> <Boolean xmlns="http://schemas.mic rosoft.com/2003/10/ Serialization/">true</bo olean> Bool true: success false: failure</pre>

4.2.26. PlaceCutStudy Service Call

POST

```
/dms/services/PlaceCutStudy
```

Places a cut in the study mode.

You can place different cut types. For example, CutType 1 is for a regular cut and CutType 8 is for a predicted loss of voltage cut.

Sample Payload	Return
<pre>{ "CutName": "TestCut", "CutId": "213ACCE0-0C43-4B04-923C-F67460CFA084", "StationName": "RAVEN", "CompId": "9DA3C978-5DE1-4960-8B0B-56F5F3BA80EF", "CompType": 5, // Transformer "FrLength": 0, "Phases": 7, // ABC "GeoName": "RAVEN", "X": 9120, "Y": -5040, "Comment": "Cut in study mode!", "CutType": 8, // PredictedCut (Loss Of Voltage) "IsClosed": true, "LblPosition": "NW" }</pre>	<pre>{ "Status": true, "Message": "String content" }</pre>

4.2.27. PlaceFaultRT Service Call

POST

/dms/services/PlaceFaultRT

Places a fault in the real-time mode.

Sample Payload	Return
<pre><PlaceTemporaryEventData xmlns="http://com.alstom.dms.rest"> <AscFault>true</AscFault> <Comment>String content</Comment> <ComponentID>String content</ComponentID> <ComponentType>2147483647</ComponentType> <CutLength>1.26743233E+15</CutLength> <DevicesToSkip>2147483647</DevicesToSkip> <FaultID>String content</FaultID> <FaultName>String content</FaultName> <GeoName>String content</GeoName> <IsMomentary>true</IsMomentary> <IsTemporary>true</IsTemporary> <Phases>String content</Phases> <Reason>2147483647</Reason> <SkipProtectiveDevices>true</SkipProtectiveDevices> <StationName>String content</StationName> <TimeToLastInSec>2147483647</TimeToLastInSec> <TripUpstreamDevice>true</TripUpstreamDevice> <UseDifferentialRelay>true</UseDifferentialRelay> <WithRecloser>true</WithRecloser> </PlaceTemporaryEventData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.28. PlaceFaultSIM Service Call

POST

/dms/services/PlaceFaultSIM

Places a fault in the simulator mode.

Sample Payload	Return
The PlaceFaultSIM call uses the PlaceTemporaryEventData schema. For the request example, see PlaceFaultRT Service Call .	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.29. PlaceJumper Service Call

POST

/dms/services/PlaceJumper

Places a jumper in the real-time mode.

Sample Payload	Return
<pre><PlaceJumperEventData xmlns="http://com.alstom.dms.rest"> <Comment>String content</Comment> <FirstCutLength>1.26743233E+15</FirstCutLength> <FirstLineID>String content</FirstLineID> <GeoName>String content</GeoName> <ID>String content</ID> <Name>String content</Name> <SecondCutLength>1.26743233E+15</SecondCutLength> <SecondLineID>String content</SecondLineID> <StationNameOfFirstLine>String content</StationNameOfFirstLine> <StationNameOfSecondLine>String content</StationNameOfSecondLine> <Status>2147483647</Status> <TransferPhasingData> <AToA>true</AToA> <AToB>true</AToB> <AToC>true</AToC> <AToN>true</AToN> <BToA>true</BToA> <BToB>true</BToB> <BToC>true</BToC> <BToN>true</BToN> <CToA>true</CToA> <CToB>true</CToB> <CToC>true</CToC> <CToN>true</CToN> <NToA>true</NToA> <NToB>true</NToB> <NToC>true</NToC> <NToN>true</NToN> </TransferPhasingData> </PlaceJumperEventData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.30. PlaceMobileSubStation Service Call

POST

/dms/services/PlaceMobileSubStation

Places a mobile substation.

Sample Payload	Return
The PlaceMobileSubStation call uses the MobileSubStationEventData schema. For the request example, see MoveMobileSubStation Service Call .	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.31. PlaceRemoveTagST Service Call

POST

/dms/services/PlaceRemoveTagST

Creates, modifies, or deletes a tag in study mode.

Sample Payload	Return
<pre>{ "Comment": "Tag created from API", "DeviceId": "22240683", "Remove": false, "StationName": "3LIONS", "TagType": "DO NOT OPERATE", "TagUser": "DIS" }</pre>	<pre>{ "Status": true, "Message": "Request was successfully sent to the DMS Server" }</pre>

4.2.32. PlaceSwitch Service Call

POST

/dms/services/PlaceSwitch

Places a temporary switch in the real-time mode.

Sample Payload	Return
The PlaceSwitch call uses the PlaceCutEventData schema. For the request example, see PlaceCut Service Call .	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.33. PlaceTemporaryGround Service Call

POST

/dms/services/PlaceTemporaryGround

Places a temporary ground.

Sample Payload	Return
The PlaceTemporaryGround call uses the PlaceTemporaryEventData schema. For the request example, see PlaceFaultRT Service Call .	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.34. RemoveCut Service Call

POST

/dms/services/RemoveCut

Removes a cut in the real-time mode.

Sample Payload	Return
<pre><RemoveCutEventData xmlns="http://com.alstom.dms.rest"> <CutID>String content</CutID> <StationName>String content</StationName> </RemoveCutEventData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.35. RemoveJumper Service Call

POST

/dms/services/RemoveJumper

Sample Payload	Return
<pre><RemoveJumperEventData xmlns="http://com.alstom.dms.rest"> <JumperID>String content</JumperID> <StationName>String content</StationName> </RemoveJumperEventData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.36. RemoveMobileSubStation Service Call

POST

```
/dms/services/RemoveMobileSubStation
```

Removes a mobile substation.

Sample Payload	Return
The RemoveMobileSubStation call uses the MobileSubStationEventData schema. For the request example, see MoveMobileSubStation Service Call .	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.37. RemoveTemporaryEventRT Service Call

POST

```
/dms/services/RemoveTemporaryEventRT
```

Removes a fault or temporary ground in the real-time mode.

Sample Payload	Return
<pre><RemoveTemporaryEventData xmlns="http://com.alstom.dms.rest"> <FaultID>String content</FaultID> <StationName>String content</StationName> </RemoveTemporaryEventData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.38. RemoveTemporaryEventSIM Service Call

POST

```
/dms/services/RemoveTemporaryEventSIM
```

Removes a fault or temporary ground in the simulator mode.

Sample Payload	Return
The RemoveTemporaryEventSIM call uses the RemoveTemporaryEventData schema. For the request example, see RemoveJumper Service Call .	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.39. RequestResolveQuery Service Call

POST

```
/dms/services/RequestResolveQuery
```

Sends a DMS query.

For example, to query all switch IDs in the Raven station, use the following request body:

```
{"QueryString":"select ID from RT.Sub(Raven).SWT"}
```

Sample Payload	Return
<pre><RequestQueryData xmlns="http://com.alstom.dms.rest"> <QueryString>String content</QueryString> </RequestQueryData></pre>	<pre><string xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">String content</string></pre>

4.2.40. ResetStudyServer Service Call

POST

```
/dms/services/ResetStudyServer
```

Resets the Study server; this clears all dynamics for the Study server.

Sample Payload	Return
The Request body is empty.	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.41. ResolveQuery Service Call

GET

```
/dms/services/ResolveQuery/{QUERY}
```

Sends a query to the DNAF server for DMS data. If the DNAF server is not available, the query is routed to the DMS server.

For example, to query all switch IDs in the Raven station, use the following url:

```
http://localhost:8801/dms/services/ResolveQuery/select ID from RT.Sub(Raven).SWT
```

Sample Payload	Return
The Request body is empty.	<pre><string xmlns="http://schemas.microsoft.com/2003/10/Serialization/">String content</string></pre>

4.2.42. SaveStudyCase Service Call

POST

/dms/services/SaveStudyCase

Creates a Study server savecase.

Note

The Study server must be running and the study environment must be initialized.

Sample Payload	Return
<pre>{ "SavecaseName": "Save15", "Comment": "", "CreatedByUserId": "212811169", "DirectoryToSave": "C:\\Eterra\\Distribution\\CurrentDev\\FantasyIsland\\ dynamics\\Client\\StudyServer\\Savecases\\", "IncludeDynamics": true, "IncludeMobileStations": true, "IncludeManuals": true, "IncludeAnnotations": true, "IncludeScadaMeasurements": true, "IncludeDnafParameters": true, "IncludedDnafSolutionData": true, "IncludedMeterMeasurements": true, "IncludedDynSchedules": true, "IncludedDerGatewayMeasurements": true }</pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.43. SendDataEntry Service Call

POST

/dms/services/SendDataEntry

Updates DNAF data. This call is used for internal purposes.

Sample Payload	Return
<pre><DataEntry xmlns="http://com.alstom.dms.rest"> <Key>SUB=COHO</Key> <SendToDnaf>false</SendToDnaf> <FieldList> <DeviceFieldEntry> <FieldName>MAN_FDR_LOADSCALING</FieldName> <NewValue>2</NewValue> </DeviceFieldEntry> </FieldList> </DataEntry></pre>	The Response body is empty.

4.2.44. SendDataEntrySim Service Call

POST

/dms/services/SendDataEntrySim

Updates DNAF data in the simulator mode. This call is used for internal purposes.

Sample Payload	Return
The SendDataEntrySim call uses the DataEntry schema. For the request example, see SendDataEntry Service Call .	The Response body is empty.

4.2.45. SendDnafPluginDisplayRequest Service Call

POST

/dms/services/SendDnafPluginDisplayRequest

Sends an LVM System Monitor configuration request to the DNAF Display plugin. This call is used for internal purposes.

Sample Payload	Return
<pre><SendDnafPluginDisplayRequest xmlns="http://com.alstom.dms.rest"> <Command>String content</Command> <Option>String content</Option> <ReadOnly>true</ReadOnly> </SendDnafPluginDisplayRequest></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.46. SetDeviceState Service Call

POST

/dms/services/SetDeviceState

Used to update device state for a distribution transformer, cut, switching device, or capacitor.

- i* **Note**

The DeviceId must exist in the DMS server.
- i* **Note**

The DeviceType value must follow the associated DeviceType coded translation item key.

Sample Payload	Return
<pre><DeviceState xmlns="http://com.alstom.dms.rest"> <StationName>Raven</StationName> <DeviceId>1234</DeviceId> <Status>true</Status> //true for CLOSE <DeviceType>4</DeviceType> <Phases>ABC</Phases> <Properties></Properties> <isConfirmDeviceState>False</isConfirmDeviceState> </DeviceState></pre>	<pre><ConfirmResponse xmlns="http://com.alstom.dms.rest"> <Status>true</Status> <ErrorMessage>String content</ErrorMessage> </ConfirmResponse></pre>

4.2.47. StudySavecaseRestore Service Call

POST

/dms/services/StudySavecaseRestore

Retrieves the specified savecase in study mode.

Note

The netconf_dataservice.xml file must include configurations for the study mode.

Sample Payload	Return
<pre><StudySaveCaseRestoreData xmlns="http://com.alstom.dms.rest"> <Path>String content</Path> <SaveCaseName>String content</SaveCaseName> </StudySaveCaseRestoreData></pre>	<pre><Boolean xmlns="http://schemas.microsoft.com/2003/10/ Serialization/">true</boolean> Bool true: success false: failure</pre>

4.2.48. SwitchStatus Service Call

POST

/dms/services/SwitchStatus

Used to update switch status. 'false': Open 'true': Closed

Sample Payload	Return
<pre><?xml version="1.0" encoding="UTF-8" standalone="yes"?> <SwitchStatus xmlns="http://com.alstom.dms.rest"> <StationName>PEBBLE BEACH</StationName> <SwitchId>22233733</SwitchId> <Status>false</Status> </SwitchStatus></pre>	None

4.2.49. TapPosition Service Call

POST

```
/dms/services/TapPosition
```

Used to update tap position of a tap changing transformer.

Sample Payload	Return
<pre>{ "StationName": "3LIONS", "DeviceId": "PTAP32E5F2FF-B2CB-47A0-8EB9-3EDD1FD9A209", "TapPosition": -1 }</pre>	<pre>{ "Status": true, "Message": "String content" }</pre>

4.2.50. TempSwitchStatus Service Call

POST

```
/dms/services/TempSwitchStatus
```

Used to update status of a temporary switch (jumper). 'false': Open 'true': Closed

Sample Payload	Return
<pre><SwitchStatus xmlns="http://com.alstom.dms.rest"> <StationName>String content</StationName> <SwitchId>String content</SwitchId> <Status>true</Status> </SwitchStatus></pre>	None

4.2.51. UpdateSwitchStatusInSim Service Call

POST

```
/dms/services/UpdateSwitchStatusInSim
```

Updates the switching device status in the simulator mode. 'false': Open 'true': Closed

Sample Payload	Return
<pre><SwitchStatus xmlns="http://com.alstom.dms.rest"> <StationName>String content</StationName> <SwitchId>String content</SwitchId> <Status>true</Status> </SwitchStatus></pre>	None

Two notification REST calls notify external applications about the status of the request. You can specify the notification format to be either XML or JSON for different content format. For more

information, see [DMS Interface Server Configuration](#).

i Note

For the applications to receive notifications, you must configure the `<NotificationBaseUrl>` and `<NotificationTypes>` in the `DmsInterfaceServerConfig.xml` file.

4.2.52. SwitchStatus Notification Call

POST

```
/<NotificationBaseUrl>/switchStatus
```

Sends notifications about switch status changes to an external system. 'false': Open 'true': Close

Sample Payload	Return
<pre><?xml version="1.0" encoding="utf-8"?> <SwitchEvent xmlns="http://com.alstom.dms.rest"> <StationName>PEBBLE BEACH</StationName> <SwitchId>22233733</SwitchId> <PendingPermanent>true</PendingPermanent> <Status>true</Status> </SwitchEvent></pre>	None

4.2.53. dnafStatus Notification Call

POST

```
/<NotificationBaseUrl>/dnafStatus
```

Sends notifications about the `DpfRequest` or `DpfStudyRequest` results.

The GUID will match the GUID that was returned in the `DpfRequest` (or `DpfStudyRequest`)

The `SensibilityList`, `VoltageViolationList`, and `OverloadViolationList` can be empty, depending on the DPF result.

Sample Payload	Return
<pre> <?xml version="1.0" encoding="utf-8"?> <DnafEvent xmlns="http://com.alstom.dms.rest"> <StationName>PEBBLE BEACH</StationName> <Guid>65f8e537-4db2-4176-8152-c8310dbd06f3</Guid> <Status>DNAF_R_SOLVED</Status> <SensibilityList> <SensitivityResult> <ControlMRID>LOAD_22191852XRESB</ControlMRID> <ViolationMRID>148990634-2_ALL_PHASES</ViolationMRID> <Sensitivity>0</Sensitivity> </SensitivityResult> ... </SensibilityList> <VoltageViolationList> <VoltageViolationOutput> <MRID>22191852</MRID> <VoltageValue>111.00</VoltageValue> <LimitValue>120.00</LimitValue> </VoltageViolationOutput> ... </VoltageViolationList> <OverloadViolationList> <OverloadViolationOutput> <MRID>143595609</MRID> <FlowValue>161.79</FlowValue> <LimitValue>100.00</LimitValue> </OverloadViolationOutput> ... </OverloadViolationList> </DnafEvent> </pre>	None

4.2.54. Annotation Service Call

POST/PUT

/dms/services/annotations

This call can be used to create and update annotations.

The ID is the key. If an annotation with the given ID does not exist in the system, it will be created. Otherwise, the existing annotation will be updated with the provided data.

Note

The annotation type cannot be updated once the annotation has been created.

Sample Payload	Return
<pre><?xml version="1.0" encoding="UTF-8" standalone="yes"?> <Annotation xmlns="http://www.esca.com/idms" xmlns:i="http://www.w3.org/2001/XMLSchema-instance"> <Id>aabbccdd</Id> <Name>FromFiddler</Name> <Type>2</Type> <Status>0</Status> <Latitude>26.57395111132735</Latitude> <Longitude>-80.13555256390715</Longitude> <StationName>PBLBEACH</StationName> <ServiceCenter>WST</ServiceCenter> <CreationTime>2014-10-10T15:31:18</CreationTime> <UpdateTime>2014-10-09T15:31:18</UpdateTime> <CreatedBy>Fiddler</CreatedBy> <UpdatedBy>Fiddler2</UpdatedBy> <Comment>test</Comment> </Annotation></pre>	None

GET

```
/dms/services/annotations/{id}
```

For getting annotation details from DMS

Sample Payload	Return
None	Annotation in XML format

DELETE

```
/dms/services/annotations/{id}
```

For deleting annotations from DMS

Sample Payload	Return
None	None

GET

```
/dms/services/annotations/lastUpdateTime?Type={type}
```

For getting the last update time for a given annotation type. If the type is empty, the system will return the last update time for all annotation types in DMS.

Sample Payload	Returns
None	Datetime

5. Switching Order Web Service

The Switching Order web service allows external applications to update switching orders. Similar to the DMS Control service (see [DMS Data Service](#)), the Switching Order web service is implemented as a set of REST POST calls. Service calls support both XML and JSON formats and can be accessed through <http://localhost:8801/dms/SWO/>.

The following types of calls are currently supported:

- Create a switching order from a work request.
- Add a switching step to a switching order.
- Add multiple switching steps to a switch order.
- Add or remove an employee from the dispatching list (requires integration with the the Mobile application).
- Get switching order details (including steps and audit trails).
- Get switching order step employee dispatch statuses (requires integration with the Mobile application).
- Get a list of locked switching order.
- Unlock a switching order.
- Query switching orders from the DMS server.
- Update switching order properties.
- Update switching order step employee dispatch statuses (requires integration with the Mobile application).
- Update switching order customer list (affected customers or traced customers).
- Update switching order status.
- Update switching order status by switching order name.
- Update switching order step status.
- Update switching order step properties.
- Remove a switching order step.
- Link/unlink a switching order from an incident.

External applications can also register for event notification about switching order status and step changes. These events are sent via a REST call to the registered URLs.

The following notification events are currently supported:

- Switching order status update notification.
- Switching order step complete notification.
- Switching order PDF document saved notification.

5.1. Service API

This section provides details about service calls. For more information, refer to <http://localhost:8801/dms/SWO/help>.

5.1.1. Add Work Request (Create a Switching Order)

HTTP Method	POST
URI	/SWO/SwitchingOrders
Response	Success: • 200: Payload contains newly created switching order. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

XML schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="WorkRequest">
    <xs:sequence>
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="FeederName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DivisionId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OperatingCenterId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="IsPlanned" type="xs:boolean" />
      <xs:element minOccurs="0" name="ScheduledStartTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ScheduledEndTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ReferenceNumber" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SearchKeyWord" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
      <xs:element minOccurs="0" name="IsHotlineHold" type="xs:boolean" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="WorkRequest" nillable="true" type="tns:WorkRequest" />
  <xs:complexType name="ArrayOfCustomField">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
```

```

type="tns:CustomField" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomField" nillable="true" type="tns:ArrayOfCustomField" />
<xs:complexType name="CustomField">
  <xs:sequence>
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="CustomField" nillable="true" type="tns:CustomField" />
</xs:schema>

```

5.1.2. Add Work Request (Create a Switching Order) Version 2

HTTP Method	POST
URI	/SWO/SwitchingOrders/WorkRequestV2
Response	Success: • 200: Payload contains newly created switching order. Failure: • 401: Unauthorized • 400: Bad Request (Invalid Parameters) • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

XML schema (request):

WorkRequestV2 Request Schema:

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
<xs:complexType name="WorkRequestV2">
<xs:complexContent mixed="false">
<xs:extension base="tns:WorkRequest">
<xs:sequence>
<xs:element minOccurs="0" name="DocumentType" type="xs:int" />
<xs:element minOccurs="0" name="IsolationDevices" nillable="true" type="tns:ArrayOfIsolationDevice" />
<xs:element minOccurs="0" name="SafetyInformation" nillable="true" type="tns:SafetyInfo" />
<xs:element minOccurs="0" name="TempGroundDevices" nillable="true"
xmlns:q1="http://schemas.datacontract.org/2004/07/NetUtil" type="q1:ArrayOfTempGroundDevice" />
<xs:element minOccurs="0" name="AdditionalData" nillable="true"
xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil" type="q2:ArrayOfPropertyUpdate" />
<xs:element minOccurs="0" name="CreateAutomaticPlan" type="xs:boolean" />

```

```

<xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Employees" nillable="true" type="tns:ArrayOfEmployeeInfo" />
<xs:element minOccurs="0" name="IsolatingDevicesDetails" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NotifyOnApprove" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NotifyOnComplete" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NotifyOnDelete" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NotifyOnExecute" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NotifyOnReject" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NotifyOnValidate" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Subtype" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="TempGroundsDetails" nillable="true" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name="WorkRequestV2" nillable="true" type="tns:WorkRequestV2" />
<xs:complexType name="WorkRequest">
<xs:sequence>
<xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="FeederName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DivisionId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OperatingCenterId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IsPlanned" type="xs:boolean" />
<xs:element minOccurs="0" name="ScheduledStartTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="ScheduledEndTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ReferenceNumber" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SearchKeyWord" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
<xs:element minOccurs="0" name="IsHotLineHold" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
<xs:element name="WorkRequest" nillable="true" type="tns:WorkRequest" />
<xs:complexType name="ArrayOfCustomField">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="tns:CustomField" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomField" nillable="true" type="tns:ArrayOfCustomField" />
<xs:complexType name="CustomField">
<xs:sequence>
<xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="CustomField" nillable="true" type="tns:CustomField" />
<xs:complexType name="ArrayOfIsolationDevice">
<xs:sequence>

```

```

<xs:element minOccurs="0" maxOccurs="unbounded" name="IsolationDevice" nillable="true"
type="tns:IsolationDevice" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfIsolationDevice" nillable="true" type="tns:ArrayOfIsolationDevice" />
<xs:complexType name="IsolationDevice">
<xs:sequence>
<xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="IsolationDevice" nillable="true" type="tns:IsolationDevice" />
<xs:complexType name="SafetyInfo">
<xs:sequence>
<xs:element minOccurs="0" name="SafetyDocHolder1" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocHolder2" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocName1" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocName2" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocSecondHolder1" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocSecondHolder2" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocType1" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyDocType2" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SafetyInfo" nillable="true" type="tns:SafetyInfo" />
<xs:complexType name="ArrayOfEmployeeInfo">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="EmployeeInfo" nillable="true"
type="tns:EmployeeInfo" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfEmployeeInfo" nillable="true" type="tns:ArrayOfEmployeeInfo" />
<xs:complexType name="EmployeeInfo">
<xs:sequence>
<xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="EmployeeInfo" nillable="true" type="tns:EmployeeInfo" />
</xs:schema>

```

XML schema (response):

The response schema is `SwitchingOrder`. For more details, see [Get Switching Order Details by WorkRequestId](#).

5.1.3. Add a Switching Step

HTTP Method	POST
-------------	------

URI	/SWO/SwitchingOrders/AddStep
Response	Success: • 200: Payload contains newly created switching order. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

XML schema (request):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
<xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
<xs:import namespace="http://com.alstom.dms.rest" />
<xs:complexType name="SwitchingOrderAddStep">
<xs:sequence>
<xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Step" nillable="true" type="tns:SwitchingOrderStep" />
<xs:element minOccurs="0" name="Feeder" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel1Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel2Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderAddStep" nillable="true" type="tns:SwitchingOrderAddStep" />
<xs:complexType name="SwitchingOrderStep">
<xs:sequence>
<xs:element minOccurs="0" name="ActualTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CancelledTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CompletedTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CompositeKey" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewETA" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
<xs:element minOccurs="0" name="DeenergizedCustomerList" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
<xs:element minOccurs="0" name="DeenergizedCustomers" type="xs:int" />
<xs:element minOccurs="0" name="DeviceID" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceState" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceStationName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceTypeName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceX" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceY" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DynamicDeviceMetadata" nillable="true" type="tns:DynamicDeviceInfo" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="0" name="EnergizedCustomerList" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfstring" />
<xs:element minOccurs="0" name="EnergizedCustomers" type="xs:int" />
<xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IsDeEnergizing" type="xs:boolean" />
<xs:element minOccurs="0" name="IsEnergizing" type="xs:boolean" />
<xs:element minOccurs="0" name="IsScadaPointControllable" type="xs:boolean" />
<xs:element minOccurs="0" name="IssuedBy" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IssuedTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="IssuedTo" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IssuedToId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Mode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="PhaseIndicator" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyInstruction" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ScheduledTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="StepInstruction" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="StepNumber" type="xs:int" />
<xs:element minOccurs="0" name="StepNumberStr" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="TagState" xmlns:q3="http://schemas.datacontract.org/2004/07/NetUtil"
type="q3:EExpectedTagState" />
<xs:element minOccurs="0" name="TagType" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStep" nillable="true" type="tns:SwitchingOrderStep" />
<xs:complexType name="ArrayOfCustomField">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="tns:CustomField" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomField" nillable="true" type="tns:ArrayOfCustomField" />
<xs:complexType name="CustomField">
<xs:sequence>
<xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="CustomField" nillable="true" type="tns:CustomField" />
<xs:complexType name="DynamicDeviceInfo">
<xs:sequence>
<xs:element name="DeviceType" type="tns:DynamicDeviceType" />
<xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CutType" type="xs:int" />
<xs:element minOccurs="0" name="TempGroundType" type="xs:int" />
<xs:element minOccurs="0" name="CutData" nillable="true" xmlns:q4="http://com.alstom.dms.rest"
type="q4:PlaceCutEventData" />
<xs:element minOccurs="0" name="JumperData" nillable="true" xmlns:q5="http://com.alstom.dms.rest"
type="q5:PlaceJumperEventData" />
<xs:element minOccurs="0" name="TemporaryGroundData" nillable="true"
xmlns:q6="http://com.alstom.dms.rest" type="q6:PlaceTemporaryEventData" />
<xs:element minOccurs="0" name="CreatedTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="CreatedBy" nillable="true" type="xs:string" />

```

```

</xs:sequence>
</xs:complexType>
<xs:element name="DynamicDeviceInfo" nillable="true" type="tns:DynamicDeviceInfo" />
<xs:simpleType name="DynamicDeviceType">
<xs:restriction base="xs:string">
<xs:enumeration value="None" />
<xs:enumeration value="Cut" />
<xs:enumeration value="Jumper" />
<xs:enumeration value="TemporaryGround" />
</xs:restriction>
</xs:simpleType>
<xs:element name="DynamicDeviceType" nillable="true" type="tns:DynamicDeviceType" />
</xs:schema>

```

XML schema (response):

The response body is empty.

5.1.4. Add a Switching Step Version 2

HTTP Method	POST
URI	/SWO/SwitchingOrders/AddStepV2
Response	Success: • 200: The payload includes user-defined responses such as request status, device ID, step ID, and dynamic device data. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

XML schema (request):

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
<xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
<xs:import namespace="http://com.alstom.dms.rest" />
<xs:complexType name="SwitchingOrderAddStepV2">
<xs:complexContent mixed="false">
<xs:extension base="tns:SwitchingOrderAddStep">
<xs:sequence>
<xs:element minOccurs="0" name="DynamicDevice" nillable="true" type="tns:DynamicDeviceInfo" />
</xs:sequence>
</xs:extension>
</xs:complexContent>

```



```

</xs:complexType>
<xs:element name="SwitchingOrderAddStepV2" nillable="true" type="tns:SwitchingOrderAddStepV2" />
<xs:complexType name="SwitchingOrderAddStep">
<xs:sequence>
<xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Step" nillable="true" type="tns:SwitchingOrderStep" />
<xs:element minOccurs="0" name="Feeder" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel1Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel2Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderAddStep" nillable="true" type="tns:SwitchingOrderAddStep" />
<xs:complexType name="SwitchingOrderStep">
<xs:sequence>
<xs:element minOccurs="0" name="ActualTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CancelledTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CompletedTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CompositeKey" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewETA" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
<xs:element minOccurs="0" name="DeenergizedCustomerList" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
<xs:element minOccurs="0" name="DeenergizedCustomers" type="xs:int" />
<xs:element minOccurs="0" name="DeviceID" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceState" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceStationName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceTypeName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceX" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceY" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DynamicDeviceMetadata" nillable="true" type="tns:DynamicDeviceInfo" />
<xs:element minOccurs="0" name="EnergizedCustomerList" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfstring" />
<xs:element minOccurs="0" name="EnergizedCustomers" type="xs:int" />
<xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IsDeEnergizing" type="xs:boolean" />
<xs:element minOccurs="0" name="IsEnergizing" type="xs:boolean" />
<xs:element minOccurs="0" name="IsScadaPointControllable" type="xs:boolean" />
<xs:element minOccurs="0" name="IssuedBy" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IssuedTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="IssuedTo" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IssuedToId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Mode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="PhaseIndicator" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyInstruction" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ScheduledTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="StepInstruction" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="StepNumber" type="xs:int" />
<xs:element minOccurs="0" name="StepNumberStr" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="TagState" xmlns:q3="http://schemas.datacontract.org/2004/07/NetUtil"

```

```

type="q3:EExpectedTagState" />
<xs:element minOccurs="0" name="TagType" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStep" nillable="true" type="tns:SwitchingOrderStep" />
<xs:complexType name="ArrayOfCustomField">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="tns:CustomField" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomField" nillable="true" type="tns:ArrayOfCustomField" />
<xs:complexType name="CustomField">
<xs:sequence>
<xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="CustomField" nillable="true" type="tns:CustomField" />
<xs:complexType name="DynamicDeviceInfo">
<xs:sequence>
<xs:element name="DeviceType" type="tns:DynamicDeviceType" />
<xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CutType" type="xs:int" />
<xs:element minOccurs="0" name="TempGroundType" type="xs:int" />
<xs:element minOccurs="0" name="CutData" nillable="true" xmlns:q4="http://com.alstom.dms.rest"
type="q4:PlaceCutEventData" />
<xs:element minOccurs="0" name="JumperData" nillable="true" xmlns:q5="http://com.alstom.dms.rest"
type="q5:PlaceJumperEventData" />
<xs:element minOccurs="0" name="TemporaryGroundData" nillable="true"
xmlns:q6="http://com.alstom.dms.rest" type="q6:PlaceTemporaryEventData" />
<xs:element minOccurs="0" name="CreatedTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="CreatedBy" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="DynamicDeviceInfo" nillable="true" type="tns:DynamicDeviceInfo" />
<xs:simpleType name="DynamicDeviceType">
<xs:restriction base="xs:string">
<xs:enumeration value="None" />
<xs:enumeration value="Cut" />
<xs:enumeration value="Jumper" />
<xs:enumeration value="TemporaryGround" />
</xs:restriction>
</xs:simpleType>
<xs:element name="DynamicDeviceType" nillable="true" type="tns:DynamicDeviceType" />
</xs:schema>

```

XML schema (response):

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="AddStepResponseV2">

```

```

<xs:sequence>
  <xs:element minOccurs="0" name="Success" type="xs:boolean" />
  <xs:element minOccurs="0" name="StepId" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="HasDynamicDeviceMetadata" type="xs:boolean" />
  <xs:element minOccurs="0" name="DynamicDeviceType" type="tns:DynamicDeviceType" />
</xs:sequence>
</xs:complexType>
<xs:element name="AddStepResponseV2" nillable="true" type="tns:AddStepResponseV2" />
<xs:simpleType name="DynamicDeviceType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="None" />
    <xs:enumeration value="Cut" />
    <xs:enumeration value="Jumper" />
    <xs:enumeration value="TemporaryGround" />
  </xs:restriction>
</xs:simpleType>
<xs:element name="DynamicDeviceType" nillable="true" type="tns:DynamicDeviceType" />
</xs:schema>

```

5.1.5. Add Switching Steps

HTTP Method	POST
URI	/SWO/SwitchingOrders/AddSteps
Response	Success: • 200: The payload includes user-defined responses such as request status and error message. Failure: • 400: Bad Request (Invalid Parameter) • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

XML schema (request):

```

<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
  <xs:import namespace="http://com.alstom.dms.rest" />
  <xs:complexType name="ArrayOfSwitchingOrderAddStepV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderAddStepV2" nillable="true"
type="tns:SwitchingOrderAddStepV2" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:complexType>
<xs:element name="ArrayOfSwitchingOrderAddStepV2" nillable="true"
type="tns:ArrayOfSwitchingOrderAddStepV2" />
<xs:complexType name="SwitchingOrderAddStepV2">
<xs:complexContent mixed="false">
<xs:extension base="tns:SwitchingOrderAddStep">
<xs:sequence>
<xs:element minOccurs="0" name="DynamicDevice" nillable="true" type="tns:DynamicDeviceInfo" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name="SwitchingOrderAddStepV2" nillable="true" type="tns:SwitchingOrderAddStepV2" />
<xs:complexType name="SwitchingOrderAddStep">
<xs:sequence>
<xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Step" nillable="true" type="tns:SwitchingOrderStep" />
<xs:element minOccurs="0" name="Feeder" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel1Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel2Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderAddStep" nillable="true" type="tns:SwitchingOrderAddStep" />
<xs:complexType name="SwitchingOrderStep">
<xs:sequence>
<xs:element minOccurs="0" name="ActualTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CancelledTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CompletedTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CompositeKey" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewETA" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
<xs:element minOccurs="0" name="DeenergizedCustomerList" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
<xs:element minOccurs="0" name="DeenergizedCustomers" type="xs:int" />
<xs:element minOccurs="0" name="DeviceID" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceState" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceStationName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceTypeName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceX" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceY" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DynamicDeviceMetadata" nillable="true" type="tns:DynamicDeviceInfo" />
<xs:element minOccurs="0" name="EnergizedCustomerList" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfstring" />
<xs:element minOccurs="0" name="EnergizedCustomers" type="xs:int" />
<xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IsDeEnergizing" type="xs:boolean" />
<xs:element minOccurs="0" name="IsEnergizing" type="xs:boolean" />
<xs:element minOccurs="0" name="IsScadaPointControllable" type="xs:boolean" />
<xs:element minOccurs="0" name="IssuedBy" nillable="true" type="xs:string" />

```

```

<xs:element minOccurs="0" name="IssuedTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="IssuedTo" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IssuedToId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Mode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="PhaseIndicator" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SafetyInstruction" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ScheduledTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="StepInstruction" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="StepNumber" type="xs:int" />
<xs:element minOccurs="0" name="StepNumberStr" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="TagState" xmlns:q3="http://schemas.datacontract.org/2004/07/NetUtil"
type="q3:EEExpectedTagState" />
<xs:element minOccurs="0" name="TagType" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStep" nillable="true" type="tns:SwitchingOrderStep" />
<xs:complexType name="ArrayOfCustomField">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="tns:CustomField" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomField" nillable="true" type="tns:ArrayOfCustomField" />
<xs:complexType name="CustomField">
<xs:sequence>
<xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="CustomField" nillable="true" type="tns:CustomField" />
<xs:complexType name="DynamicDeviceInfo">
<xs:sequence>
<xs:element name="DeviceType" type="tns:DynamicDeviceType" />
<xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CutType" type="xs:int" />
<xs:element minOccurs="0" name="TempGroundType" type="xs:int" />
<xs:element minOccurs="0" name="CutData" nillable="true" xmlns:q4="http://com.alstom.dms.rest"
type="q4:PlaceCutEventData" />
<xs:element minOccurs="0" name="JumperData" nillable="true" xmlns:q5="http://com.alstom.dms.rest"
type="q5:PlaceJumperEventData" />
<xs:element minOccurs="0" name="TemporaryGroundData" nillable="true"
xmlns:q6="http://com.alstom.dms.rest" type="q6:PlaceTemporaryEventData" />
<xs:element minOccurs="0" name="CreatedTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="CreatedBy" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="DynamicDeviceInfo" nillable="true" type="tns:DynamicDeviceInfo" />
<xs:simpleType name="DynamicDeviceType">
<xs:restriction base="xs:string">
<xs:enumeration value="None" />
<xs:enumeration value="Cut" />
<xs:enumeration value="Jumper" />

```

```

<xs:enumeration value="TemporaryGround" />
</xs:restriction>
</xs:simpleType>
<xs:element name="DynamicDeviceType" nillable="true" type="tns:DynamicDeviceType" />
</xs:schema>

```

XML schema (response):

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:complexType name="AddStepsResponse">
<xs:sequence>
<xs:element minOccurs="0" name="Success" type="xs:boolean" />
<xs:element minOccurs="0" name="ErrorMessage" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="AddStepsResponse" nillable="true" type="tns:AddStepsResponse" />
</xs:schema>

```

5.1.6. Add or Remove an Employee from the Dispatching List

HTTP Method	POST
URI	/SWO/SwitchingOrders/AssignDispatchEmployee
Response	Success: • 200: Payload contains dispatch status for employees. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid field value (duplicate or missing)

Request example (JSON):

```

{
  "WorkRequestId": "DMSALL_AUTO_20210517AA",
  "EmployeeId": "FI1607",
  "Delete": false
},
{
  "WorkRequestId": "DMSALL_AUTO_20210517AA",
  "EmployeeId": "FI1509",
  "Delete": true
}

```

Response example (JSON):

```

{
  "EmployeeId": "FI1607",

```

```

"StatusCode":0,
"StatusMessage":"Add operation successful"
},
{
"EmployeeId":"FI1509",
"StatusCode":0,
"StatusMessage":"Remove operation successful"
}

```

XML request schema:

```

<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfDispatchEmployee">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="DispatchEmployee" nillable="true"
type="tns:DispatchEmployee" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfDispatchEmployee" nillable="true" type="tns:ArrayOfDispatchEmployee" />
  <xs:complexType name="DispatchEmployee">
    <xs:sequence>
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Delete" type="xs:boolean" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="DispatchEmployee" nillable="true" type="tns:DispatchEmployee" />
</xs:schema>

```

XML response schema:

```

<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfDispatchEmployeeResponse">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="DispatchEmployeeResponse" nillable="true"
type="tns:DispatchEmployeeResponse" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfDispatchEmployeeResponse" nillable="true"
type="tns:ArrayOfDispatchEmployeeResponse" />
  <xs:complexType name="DispatchEmployeeResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StatusCode" type="xs:int" />
    </xs:sequence>
  <!--
enum EAssignDispatchEmployeeErrorCodes
{
  None = 0,
  SwitchOrderDoesntExist,
  LockedByAnotherUser,

```

```

        CantUpdateCompletedSwitchOrder,
        CannotUpdateSwitchOrder,
        EmployeeIsInCanTakeBackDispatchState,
        EmployeeHasIssueStepAssignedToHim,
        ThereIsNoModelledEmployeesOnTheSystem
    }
-->
    <xs:element minOccurs="0" name="StatusMessage" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="DispatchEmployeeResponse" nillable="true" type="tns:DispatchEmployeeResponse" />
</xs:schema>

```

5.1.7. Get Employee Dispatch Statuses

HTTP Method	POST
URI	/SWO/SwitchingOrders/GetDispatchStatuses
Response	Success: • 200: Payload contains employee dispatch statuses. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

XML schema (request):

```

<xs:schema xmlns:tns="http://www.esca.com/ids" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="GetDispatchStatusRequest">
    <xs:sequence>
      <xs:element minOccurs="0" name="EmployeeIds" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="GetDispatchStatusRequest" nillable="true" type="tns:GetDispatchStatusRequest" />
</xs:schema>

```

XML schema (response):

```

<xs:schema xmlns:tns="http://www.esca.com/ids" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="DispatchStatusResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="MobileEmployeesData" nillable="true"
type="tns:ArrayOfMobileEmployeeData" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="DispatchStatusResponse" nillable="true" type="tns:DispatchStatusResponse" />
</xs:schema>

```



```

</xs:sequence>
</xs:complexType>
<xs:element name="DispatchStatusResponse" nillable="true" type="tns:DispatchStatusResponse" />
<xs:complexType name="ArrayOfMobileEmployeeData">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="MobileEmployeeData" nillable="true"
type="tns:MobileEmployeeData" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfMobileEmployeeData" nillable="true" type="tns:ArrayOfMobileEmployeeData" />
<xs:complexType name="MobileEmployeeData">
  <xs:sequence>
    <xs:element minOccurs="0" name="Device" nillable="true" type="tns:MobileDevice" />
    <xs:element minOccurs="0" name="DispatchInfo" nillable="true" type="tns:DispatchData" />
    <xs:element minOccurs="0" name="Employee" nillable="true" type="tns:MobileEmployee" />
    <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="MobileEmployeeData" nillable="true" type="tns:MobileEmployeeData" />
<xs:complexType name="MobileDevice">
  <xs:sequence>
    <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Online" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
<xs:element name="MobileDevice" nillable="true" type="tns:MobileDevice" />
<xs:complexType name="DispatchData">
  <xs:sequence>
    <xs:element minOccurs="0" name="DispatchState" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SWONumber" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="DispatchData" nillable="true" type="tns:DispatchData" />
<xs:complexType name="MobileEmployee">
  <xs:sequence>
    <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IsMobile" type="xs:boolean" />
    <xs:element minOccurs="0" name="LogOn" type="xs:boolean" />
    <xs:element minOccurs="0" name="Online" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
<xs:element name="MobileEmployee" nillable="true" type="tns:MobileEmployee" />
</xs:schema>

```

5.1.8. Get a List of Locked Switching Orders

HTTP Method	GET
URI	/SWO/SwitchingOrders/LockedList

Response	Success: • 200: Payload contains the list of locked switching orders. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (invalid, duplicate, or missing)
----------	--

XML schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfLockedSWO">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="LockedSWO" nillable="true"
type="tns:LockedSWO" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfLockedSWO" nillable="true" type="tns:ArrayOfLockedSWO" />
  <xs:complexType name="LockedSWO">
    <xs:sequence>
      <xs:element minOccurs="0" name="SWOName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Locked" type="xs:boolean" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="LockedSWO" nillable="true" type="tns:LockedSWO" />
</xs:schema>
```

5.1.9. Unlock a Switching Order by Name

HTTP Method	GET
URI	/SWO/SwitchingOrders/Unlock/{SWONAME}
Response	Success: • 200: Payload contains the switching order unlock status. Failure: • 401: Unauthorized • 500: Generic\ Server Error

XML schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="LockedSWO">
    <xs:sequence>
      <xs:element minOccurs="0" name="SWOName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Locked" type="xs:boolean" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="LockedSWO" nillable="true" type="tns:LockedSWO" />
</xs:schema>
```

```
</xs:sequence>
</xs:complexType>
<xs:element name="LockedSWO" nillable="true" type="tns:LockedSWO" />
</xs:schema>
```

5.1.10. Unlock a Switching Order by Name or WorkRequestId

HTTP Method	POST
URI	/SWO/SwitchingOrders/Unlock/
Response	Success: • 200: Payload contains the switching order unlock status. Failure: • 401: Unauthorized • 500: Generic\ Server Error

XML schema (request):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="UnlockSwitchingOrder">
    <xs:sequence>
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="UnlockSwitchingOrder" nillable="true" type="tns:UnlockSwitchingOrder" />
</xs:schema>
```

XML schema (response):

The response schema is LockedSWO. For more details, see [Unlock a Switching Order by Name](#).

5.1.11. Query Switching Orders by Parameters

HTTP Method	GET
URI	/SWO/SwitchingOrders?Status=xxx&CreatedFrom=xxx&CreatedTo=xxx&ScheduledStartFrom=xxx& ScheduledStartTo=xxx&ScheduledEndFrom=xxx& ScheduledEndTo=xxx&LastModifiedTimeFrom=xxx& LastModifiedTimeTo=xxx&SearchKeyWord=xxx& Division=xxx&OperatingCenter=xxx
Response	Success: • 200: Payload Failure: • 500: Generic Server Error

Search Criteria	• Status (CREATED, VALIDATED, APPROVED, EXECUTING, COMPLETED, and ALL). The default value is ALL. • From/To Date (Created, ScheduledStart, ScheduledEnd, or LastModifiedTime). By default, searches for all dates. • Search keyword in SWO. By default, matches any. • Division. By default, matches any. • Operating Center. By default, matches any.
-----------------	--

XML schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfSwitchingOrderInfo">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderInfo" nillable="true"
type="tns:SwitchingOrderInfo" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfSwitchingOrderInfo" nillable="true" type="tns:ArrayOfSwitchingOrderInfo" />
  <xs:complexType name="SwitchingOrderInfo">
    <xs:sequence>
      <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DivisionId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OperatingCenterId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Status" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ScheduledStartTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ScheduledEndTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="CreatedTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="LastModifiedTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ActualStartTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ValidatedTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedTime1" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedTime2" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedTime3" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ExecutedTime" nillable="true" type="xs:dateTime" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="SwitchingOrderInfo" nillable="true" type="tns:SwitchingOrderInfo" />
</xs:schema>
```

5.1.12. Get Switching Order Details by WorkRequestId

HTTP Method	GET
URI	/SWO/SwitchingOrders/{WorkRequestId}

Response**Success:**

- 200: Payload contains the switching order details

Failure:

- 401: Unauthorized
 - 404: Not found
 - 500: Generic Server Error
-

XML schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="SwitchingOrder">
    <xs:sequence>
      <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ReferenceNumber" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="FeederName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DivisionId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OperatingCenterId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Version" type="xs:unsignedInt" />
      <xs:element minOccurs="0" name="IsPlanned" type="xs:boolean" />
      <xs:element minOccurs="0" name="Status" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ScheduledStartTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ScheduledEndTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ActualStartTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ActualEndTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="CreatedTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="LastModifiedTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="PreparedBy" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ValidatedBy" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ValidatedTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedBy1" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ApprovedBy2" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ApprovedBy3" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ApprovedTime1" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedTime2" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedTime3" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="ApprovedTo" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ExecutedBy" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="CompletedBy" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ExecutedTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SearchKeyWord" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="LastModifiedByUser" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Steps" nillable="true" type="tns:ArrayOfSwitchingOrderStep" />
      <xs:element minOccurs="0" name="AuditTrails" nillable="true"
type="tns:ArrayOfSwitchingOrderAuditTrail" />
    
```

```

    <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
    <xs:element minOccurs="0" name="SwitchOrderNumber" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Employees" nillable="true" type="tns:ArrayOfEmployeeInfo" />
    <xs:element minOccurs="0" name="SequentialNumber" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ReviewedBy1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ReviewTime1" type="xs:dateTime" />
    <xs:element minOccurs="0" name="IsolationDevices" nillable="true"
type="tns:ArrayOfIsolationDeviceEntry" />
    <xs:element minOccurs="0" name="IsolatingDevicesDetails" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="TempGroundDevices" nillable="true"
type="tns:ArrayOfTempGroundDeviceEntry" />
    <xs:element minOccurs="0" name="TempGroundsDetails" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="PlannedOutageId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="PlannedOutageName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IncidentId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IncidentName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Subtype" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DocType" type="xs:int" />
    <xs:element minOccurs="0" name="StepBlocks" nillable="true" type="tns:ArrayOfStepBlockEntry" />
    <xs:element minOccurs="0" name="DocumentLinks" nillable="true"
type="tns:ArrayOfDocumentLinkEntry" />
    <xs:element minOccurs="0" name="SafetyInformation" nillable="true" type="tns:SafetyInfo" />
    <xs:element minOccurs="0" name="NotifyOnValidate" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="NotifyOnApprove" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="NotifyOnReject" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="NotifyOnExecute" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="NotifyOnComplete" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="NotifyOnDelete" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="WorkType" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrder" nillable="true" type="tns:SwitchingOrder" />
<xs:complexType name="ArrayOfSwitchingOrderStep">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderStep" nillable="true"
type="tns:SwitchingOrderStep" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfSwitchingOrderStep" nillable="true" type="tns:ArrayOfSwitchingOrderStep" />
<xs:complexType name="SwitchingOrderStep">
  <xs:sequence>
    <xs:element minOccurs="0" name="ActualTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" name="CancelledTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CompletedTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfCustomField" />
    <xs:element minOccurs="0" name="DeviceID" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceState" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceStationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceTypeName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceX" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceY" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>

```

```

    <xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IssuedBy" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IssuedTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="IssuedTo" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IssuedToId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Mode" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyInstruction" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyInstruction" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="PhaseIndicator" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StepNumber" type="xs:int" />
    <xs:element minOccurs="0" name="StepNumberStr" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="TagType" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStep" nillable="true" type="tns:SwitchingOrderStep" />
<xs:complexType name="ArrayOfCustomField">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="tns:CustomField" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomField" nillable="true" type="tns:ArrayOfCustomField" />
<xs:complexType name="CustomField">
  <xs:sequence>
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="CustomField" nillable="true" type="tns:CustomField" />
<xs:complexType name="ArrayOfSwitchingOrderAuditTrail">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderAuditTrail" nillable="true"
type="tns:SwitchingOrderAuditTrail" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfSwitchingOrderAuditTrail" nillable="true"
type="tns:ArrayOfSwitchingOrderAuditTrail" />
<xs:complexType name="SwitchingOrderAuditTrail">
  <xs:sequence>
    <xs:element minOccurs="0" name="ModifiedbyUser" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Status" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceState" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ModifiedTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Version" type="xs:unsignedInt" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderAuditTrail" nillable="true" type="tns:SwitchingOrderAuditTrail" />
<xs:complexType name="ArrayOfEmployeeInfo">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="EmployeeInfo" nillable="true"
type="tns:EmployeeInfo" />
  </xs:sequence>

```

```

</xs:complexType>
<xs:element name="ArrayOfEmployeeInfo" nillable="true" type="tns:ArrayOfEmployeeInfo" />
<xs:complexType name="EmployeeInfo">
  <xs:sequence>
    <xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="EmployeeInfo" nillable="true" type="tns:EmployeeInfo" />
<xs:complexType name="ArrayOfIsolationDeviceEntry">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="IsolationDeviceEntry" nillable="true"
type="tns:IsolationDeviceEntry" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfIsolationDeviceEntry" nillable="true"
type="tns:ArrayOfIsolationDeviceEntry" />
<xs:complexType name="IsolationDeviceEntry">
  <xs:sequence>
    <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StructureId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StructureTypeId" type="xs:int" />
    <xs:element minOccurs="0" name="StructureName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StructureOwner" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StructureType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StructureX" type="xs:float" />
    <xs:element minOccurs="0" name="StructureY" type="xs:float" />
  </xs:sequence>
</xs:complexType>
<xs:element name="IsolationDeviceEntry" nillable="true" type="tns:IsolationDeviceEntry" />
<xs:complexType name="ArrayOfTempGroundDeviceEntry">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="TempGroundDeviceEntry" nillable="true"
type="tns:TempGroundDeviceEntry" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfTempGroundDeviceEntry" nillable="true"
type="tns:ArrayOfTempGroundDeviceEntry" />
<xs:complexType name="TempGroundDeviceEntry">
  <xs:sequence>
    <xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="PULength" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="X" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Y" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="TempGroundDeviceEntry" nillable="true" type="tns:TempGroundDeviceEntry" />
<xs:complexType name="ArrayOfStepBlockEntry">
  <xs:sequence>

```



```

    <xs:element minOccurs="0" maxOccurs="unbounded" name="StepBlockEntry" nillable="true"
type="tns:StepBlockEntry" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfStepBlockEntry" nillable="true" type="tns:ArrayOfStepBlockEntry" />
<xs:complexType name="StepBlockEntry">
  <xs:sequence>
    <xs:element minOccurs="0" name="BlockName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="BlockColor" type="xs:int" />
    <xs:element minOccurs="0" name="BlockIndexMin" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="BlockIndexMax" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="BlockDependency" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
    <xs:element minOccurs="0" name="AllowReordering" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
<xs:element name="StepBlockEntry" nillable="true" type="tns:StepBlockEntry" />
<xs:complexType name="ArrayOfDocumentLinkEntry">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="DocumentLinkEntry" nillable="true"
type="tns:DocumentLinkEntry" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfDocumentLinkEntry" nillable="true" type="tns:ArrayOfDocumentLinkEntry" />
<xs:complexType name="DocumentLinkEntry">
  <xs:sequence>
    <xs:element minOccurs="0" name="DocumentId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DocumentName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DocumentType" type="xs:int" />
  </xs:sequence>
</xs:complexType>
<xs:element name="DocumentLinkEntry" nillable="true" type="tns:DocumentLinkEntry" />
<xs:complexType name="SafetyInfo">
  <xs:sequence>
    <xs:element minOccurs="0" name="SafetyDocHolder1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocHolder2" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocName1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocName2" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocSecondHolder1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocSecondHolder2" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocType1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SafetyDocType2" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SafetyInfo" nillable="true" type="tns:SafetyInfo" />
<xs:complexType name="HotLineHold">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:SwitchingOrder">
      <xs:sequence>
        <xs:element minOccurs="0" name="EmployeeName" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="RadioChannel" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="RadioCallSign" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="PrivateNumber" nillable="true" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

```

```

    <xs:element minOccurs="0" name="CellPhoneNumber" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="PagerNumber" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="TagId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="TagCreator" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="WorkLocation" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="WorkingDesk" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ExecutionPhase" type="xs:int" />
    <xs:element minOccurs="0" name="HlhStatus" type="xs:int" />
    <xs:element minOccurs="0" name="HlhDeviceAddress" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="HlhDeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="HlhDeviceType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="HlhDeviceIsScada" type="xs:boolean" />
    <xs:element minOccurs="0" name="HlhIssuedTo" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name="HotLineHold" nillable="true" type="tns:HotLineHold" />
<xs:complexType name="PlannedRequest">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:SwitchingOrder">
      <xs:sequence>
        <xs:element minOccurs="0" name="RequestType" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="ReasonForWork" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="CausesOutage" type="xs:boolean" />
        <xs:element minOccurs="0" name="IsClearanceNeeded" type="xs:boolean" />
        <xs:element minOccurs="0" name="RequestorComments" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="PriorityNumber" type="xs:int" />
        <xs:element minOccurs="0" name="IsUrgent" type="xs:boolean" />
        <xs:element minOccurs="0" name="ReasonForUrgency" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="Address" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="CustomerNotificationMessage" nillable="true"
type="xs:string" />
        <xs:element minOccurs="0" name="NotificationMethod" type="xs:int" />
        <xs:element minOccurs="0" name="AffectsCriticalCustomers" type="xs:boolean" />
        <xs:element minOccurs="0" name="AffectedCustomersList" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfstring" />
        <xs:element minOccurs="0" name="Customers" nillable="true" type="tns:ArrayOfCustomerEntry"
/>
        <xs:element minOccurs="0" name="ReviewedBy2" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="ReviewTime2" type="xs:dateTime" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:element name="PlannedRequest" nillable="true" type="tns:PlannedRequest" />
<xs:complexType name="ArrayOfCustomerEntry">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomerEntry" nillable="true"
type="tns:CustomerEntry" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfCustomerEntry" nillable="true" type="tns:ArrayOfCustomerEntry" />

```

```

<xs:complexType name="CustomerEntry">
  <xs:sequence>
    <xs:element minOccurs="0" name="CustomerName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CustomerAddr" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Phone" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ServicePoint" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="AccountNumber" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CustomerKey" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IsCritical" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
<xs:element name="CustomerEntry" nillable="true" type="tns:CustomerEntry" />
</xs:schema>

```

5.1.13. Get Switching Order Details by Name

HTTP Method	GET
URI	/SWO/SwitchingOrdersByName/{SWITCHINGORDERNAME}
Response	Success: • 200: Payload contains the switching order details Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error

XML schema:

The response schema is `SwitchingOrder`. For more details about the schema, see [Get Switching Order Details by WorkRequestId](#).

5.1.14. Update Switching Order Properties

HTTP Method	POST
URI	/SWO/SwitchingOrders/Properties

Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 512: Invalid SWO field Name • 513: Switching Order is locked by another user • 517: Switching Order server is in read-only mode
Required Information	• Work Request ID • User Name • List of fields to be updated. Each of the fields must contain the name and the value. The fields that can be updated are: StationName, DivisionId, OperatingCenterId, FeederName, ScheduledStartTime, ActualStartTime, ScheduledEndTime, ActualEndTime, ActualEndTime, and any CustomFields

Request example (JSON):

The following example shows how to update the planned request start and end time values. This request also specifies the system to make corresponding time updates to the associated planned outage and switching order. (This functionality also requires that the OMS Interface Server is running.)

```
{
  "UserName" : "ROBOCALL",
  "Properties" :
  [
    {
      "Name" : "ScheduledStartTime",
      "Value" : "2021-10-25T19:25:43.511Z"
    },
    {
      "Name" : "ScheduledEndTime",
      "Value" : "2021-10-26T19:25:43.511Z"
    }
  ],
  "SwitchingOrderName" : "PR-21000002",
  "PropagateChangesToLinkedPO": true,
  "PropagateChangesToLinkedSwo": true
}
```

XML schema (request):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
  <xs:complexType name="SwitchingOrderPropertyUpdates">
```

```

<xs:sequence>
  <xs:element minOccurs="0" name="IsolationDevices" nillable="true"
xmlns:q1="http://schemas.datacontract.org/2004/07/NetUtil" type="q1:ArrayOfIsolationDevice" />
  <xs:element minOccurs="0" name="Properties" nillable="true"
type="tns:ArrayOfSwitchingOrderProperty" />
  <xs:element minOccurs="0" name="TempGroundDevices" nillable="true"
xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil" type="q2:ArrayOfTempGroundDevice" />
  <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="PropagateChangesToLinkedPO" type="xs:boolean" />
  <xs:element minOccurs="0" name="PropagateChangesToLinkedSwo" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderPropertyUpdates" nillable="true"
type="tns:SwitchingOrderPropertyUpdates" />
<xs:complexType name="ArrayOfSwitchingOrderProperty">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderProperty" nillable="true"
type="tns:SwitchingOrderProperty" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfSwitchingOrderProperty" nillable="true"
type="tns:ArrayOfSwitchingOrderProperty" />
<xs:complexType name="SwitchingOrderProperty">
  <xs:sequence>
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderProperty" nillable="true" type="tns:SwitchingOrderProperty" />
</xs:schema>

```

XML schema (response):

The Response body is empty.

5.1.15. Update Employee Dispatch Statuses

HTTP Method	POST
URI	/SWO/SwitchingOrders/SetDispatchStatuses

Response**Success:**

- 200: Payload contains employee dispatch statuses.

Failure:

- 401: Unauthorized
 - 500: Generic\ Server Error
 - 511: Invalid Switching Order Name (duplicate or missing)
 - 516: Invalid WorkRequestId (duplicate or missing)
 - 517: SWO server is in read-only mode
-

XML schema (request):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfSetDispatchStatusRequest">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="SetDispatchStatusRequest" nillable="true"
type="tns:SetDispatchStatusRequest" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfSetDispatchStatusRequest" nillable="true"
type="tns:ArrayOfSetDispatchStatusRequest" />
  <xs:complexType name="SetDispatchStatusRequest">
    <xs:sequence>
      <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DispatchStatus" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SWONumber" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="SetDispatchStatusRequest" nillable="true" type="tns:SetDispatchStatusRequest" />
</xs:schema>
```

XML schema (response):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="DispatchStatusResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="MobileEmployeesData" nillable="true"
type="tns:ArrayOfMobileEmployeeData" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="DispatchStatusResponse" nillable="true" type="tns:DispatchStatusResponse" />
  <xs:complexType name="ArrayOfMobileEmployeeData">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="MobileEmployeeData" nillable="true"
type="tns:MobileEmployeeData" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfMobileEmployeeData" nillable="true" type="tns:ArrayOfMobileEmployeeData" />
</xs:schema>
```

```

<xs:complexType name="MobileEmployeeData">
  <xs:sequence>
    <xs:element minOccurs="0" name="Device" nillable="true" type="tns:MobileDevice" />
    <xs:element minOccurs="0" name="DispatchInfo" nillable="true" type="tns:DispatchData" />
    <xs:element minOccurs="0" name="Employee" nillable="true" type="tns:MobileEmployee" />
    <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="MobileEmployeeData" nillable="true" type="tns:MobileEmployeeData" />
<xs:complexType name="MobileDevice">
  <xs:sequence>
    <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Online" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
<xs:element name="MobileDevice" nillable="true" type="tns:MobileDevice" />
<xs:complexType name="DispatchData">
  <xs:sequence>
    <xs:element minOccurs="0" name="DispatchState" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SWONumber" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="DispatchData" nillable="true" type="tns:DispatchData" />
<xs:complexType name="MobileEmployee">
  <xs:sequence>
    <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IsMobile" type="xs:boolean" />
    <xs:element minOccurs="0" name="LogOn" type="xs:boolean" />
    <xs:element minOccurs="0" name="Online" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
<xs:element name="MobileEmployee" nillable="true" type="tns:MobileEmployee" />
</xs:schema>

```

5.1.16. Update Switching Order Customer List

HTTP Method	POST
URI	/SWO/SwitchingOrders/SetCustomerList
Response	Success: • 200: Payload contains employee dispatch statuses. Failure: • 401: Unauthorized • 500: Generic\ Server Error • 511: Invalid Switching Order Name (duplicate or missing) • 516: Invalid WorkRequestId (duplicate or missing) • 517: SWO server is in read-only mode

JSON example (request):

The following example shows how to set a customer list for a planned request. This request also specifies the system to make corresponding customer list updates to the associated planned outage. (This functionality also requires that the OMS Interface Server is running.)

```
{
  "UserName" : "ROBOCALL",
  "SwitchingOrderName" : "PR-21000002",
  "Customers" :
  [
    { "CustomerName" : "John Doe", "CustomerAddr" : "Coho street", "Phone" : "555-123-45-67",
      "ServicePoint" : "SP_12345", "AccountNumber" : "5551234567", "CustomerKey" : "55512345675551234567",
      "IsCritical" : "false" }
    ,
    { "CustomerName" : "Mike Doe", "CustomerAddr" : "Raven avenue", "Phone" : "555-76-543-21",
      "ServicePoint" : "SP_54321", "AccountNumber" : "5557654321", "CustomerKey" : "55576543215557654321",
      "IsCritical" : "true" }
  ],
  "PropagateChangesToLinkedPO" : true
}
```

XML schema (request):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="SwitchingOrderSetCustomersListUpdate">
    <xs:sequence>
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Customers" nillable="true" type="tns:ArrayOfCustomerInfo" />
      <xs:element minOccurs="0" name="PropagateChangesToLinkedPO" type="xs:boolean" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="SwitchingOrderSetCustomersListUpdate" nillable="true"
type="tns:SwitchingOrderSetCustomersListUpdate" />
  <xs:complexType name="ArrayOfCustomerInfo">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomerInfo" nillable="true"
type="tns:CustomerInfo" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfCustomerInfo" nillable="true" type="tns:ArrayOfCustomerInfo" />
  <xs:complexType name="CustomerInfo">
    <xs:sequence>
      <xs:element minOccurs="0" name="CustomerName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="CustomerAddr" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Phone" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ServicePoint" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AccountNumber" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="CustomerInfo" nillable="true" type="tns:CustomerInfo" />

```



```

<xs:element minOccurs="0" name="CustomerKey" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IsCritical" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
<xs:element name="CustomerInfo" nillable="true" type="tns:CustomerInfo" />
</xs:schema>

```

XML schema (response):

The response body is empty.

5.1.17. Update Switching Order Status

HTTP Method	POST
URI	/SWO/SwitchingOrders/Status
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 513: Switching Order is locked by another user • 514: Invalid SWO Status (Invalid SWO status for the requested action. For example, if SWO is already in the "Executing" state, the requested action cannot be REJECT/APPROVE.) • 515: Action not supported (if the requested action is other than REJECT/APPROVE/DELETE) • 517: Switching Order server is in read-only mode

Note

The switching order workflow can be configured to exclude the review and approval stages so that the APPROVED and VALIDATED statuses are skipped and the REJECT and APPROVE actions are disabled. For more details, refer to the *Switching Order Configuration Editor User Guide*.

XML schema:

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="SwitchingOrderStatusUpdates">
    <xs:sequence>
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Action" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

```

<xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="PropagateChangesToLinkedPO" type="xs:boolean" />
<xs:element minOccurs="0" name="PropagateChangesToLinkedSwo" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStatusUpdates" nillable="true"
type="tns:SwitchingOrderStatusUpdates" />
</xs:schema>

```

5.1.18. Update Switching Order Status by Name

HTTP Method	POST
URI	/SWO/SwitchingOrders/UpdateStatusByName
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 513: Switching Order is locked by another user • 514: Invalid SWO Status (Invalid SWO status for the requested action. For example, if SWO is already in the "Executing" state, the requested action cannot be REJECT/APPROVE.) • 515: Action not supported (if the requested action is other than REJECT/APPROVE/DELETE/EXECUTE/COMPLETE) • 517: Switching Order server is in read-only mode

Request example (JSON):

The following example shows how to delete a planned request. This request also specifies that the system deletes the associated planned outage and switching order. (This functionality also requires that the OMS Interface Server is running.)

```

{
  "Action": "DELETE",
  "SwitchingOrderName": "PR-21000002",
  "UserName": "ROBOCALL",
  "PropagateChangesToLinkedPO" : true,
  "PropagateChangesToLinkedSwo" : true
}

```

XML schema:

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="SwitchingOrderStatusUpdateByName">

```

```

<xs:sequence>
  <xs:element minOccurs="0" name="Action" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="PropagateChangesToLinkedPO" type="xs:boolean" />
  <xs:element minOccurs="0" name="PropagateChangesToLinkedSwo" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStatusUpdateByName" nillable="true"
type="tns:SwitchingOrderStatusUpdateByName" />
</xs:schema>

```

5.1.19. Update Switching Order Step Status

HTTP Method	POST
URI	/SWO/SwitchingOrderStep/Status
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 513: Switching Order is locked by another user • 517: Switching Order server is in read-only mode

XML schema (request):

```

<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="SwitchingOrderStepStatusUpdates">
    <xs:sequence>
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StepId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StateName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="CompletedTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="Action" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="SwitchingOrderStepStatusUpdates" nillable="true"
type="tns:SwitchingOrderStepStatusUpdates" />

```

```
</xs:schema>
```

XML schema (response):

The response body is empty.

5.1.20. Update Switching Order Step Properties

HTTP Method	POST
URI	/SWO/SwitchingOrderStep/Properties
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 513: Switching Order is locked by another user • 517: Switching Order server is in read-only mode

Request Example (JSON):

To assign an employee to a step:

```
{
  "WorkRequestId" : "DMSALL_AUTO_20210719AA",
  "StepId" : "76a6201a-c258-4918-bef6-a88165fd4bd6",
  "Properties" : [{"Name":"IssuedTo", "Value":"John"}]
}
```

XML schema (request):

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="SwitchingOrderStepPropertyUpdate">
    <xs:sequence>
      <xs:element name="Properties" nillable="true" type="tns:ArrayOfSwitchingOrderProperty" />
      <xs:element name="StepId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="SwitchingOrderStepPropertyUpdate" nillable="true"
type="tns:SwitchingOrderStepPropertyUpdate" />
  <xs:complexType name="ArrayOfSwitchingOrderProperty">
    <xs:sequence>
```

```

    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderProperty" nillable="true"
type="tns:SwitchingOrderProperty" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfSwitchingOrderProperty" nillable="true"
type="tns:ArrayOfSwitchingOrderProperty" />
<xs:complexType name="SwitchingOrderProperty">
  <xs:sequence>
    <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderProperty" nillable="true" type="tns:SwitchingOrderProperty" />
</xs:schema>

```

XML schema (response):

The response body is empty.

5.1.21. Remove a Switching Order Step

HTTP Method	POST
URI	/SWO/SwitchingOrderStep/Remove
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 513: Switching Order is locked by another user • 517: Switching Order server is in read-only mode

Request example (XML):

To remove a step:

```

<SwitchingOrderStepRemove xmlns="http://www.esca.com/ids">
  <WorkRequestId>DMSALL_AUTO_20210729AA</WorkRequestId>
  <StepId>6096eaf5-0966-4c4e-a862-04f7f98a727f</StepId>
  <UserName>DMSALL</UserName>
</SwitchingOrderStepRemove>

```

XML schema (request):

```

<xs:schema xmlns:tns="http://www.esca.com/ids" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">

```

```

<xs:complexType name="SwitchingOrderStepRemove">
  <xs:sequence>
    <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="StepId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="SwitchingOrderStepRemove" nillable="true" type="tns:SwitchingOrderStepRemove" />
</xs:schema>

```

XML schema (response):

The response body is empty.

5.1.22. Link Switching Order to Incident

HTTP Method	POST
URI	/SWO/SwitchingOrders/LinkToIncident
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error • 513: Switching Order is locked by another user • 517: Switching Order server is in read-only mode

XML schema:

```

<xs:schema xmlns:tns="http://www.esca.com/ids" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="SwitchingOrderLinkToIncidentUpdate">
    <xs:sequence>
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SwitchingOrderName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Incident" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="SwitchingOrderLinkToIncidentUpdate" nillable="true"
type="tns:SwitchingOrderLinkToIncidentUpdate" />
</xs:schema>

```

5.1.23. Switching Order Status Change Notification

This notification is sent from the DMS interface server to the listening client.

Error Handling	Reflected in the HTTP response message.
HTTP Method	POST
URI	/SWO/StatusNotification

XML schema:

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/ids" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="SwitchingOrderStatusNotification" type="tns:SwitchingOrderStatusNotification" />
<xs:complexType name="SwitchingOrderStatusNotification">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="WorkRequestId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Version" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="PreviousStatus" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="NewStatus" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ModifiedByUser" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ModifiedTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="SwitchOrderNumber" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>
```

5.1.24. Switching Order Save to PDF Notification

This notification is sent from the DMS interface server to the listening client.

Error Handling	Reflected in HTTP response message.
HTTP Method	POST
URI	/SWO/PdfNotification

XML schema:

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/ids" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="SwitchingOrderPdfNotification" type="tns:SwitchingOrderPdfNotification" />
<xs:complexType name="SwitchingOrderPdfNotification">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="WorkRequestId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Version" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="FilePath" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>
```

```

    <xs:element minOccurs="1" maxOccurs="1" name="CreationTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="SwitchOrderNumber" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

5.1.25. Switching Order Step Completion Notification

This notification is sent from the DMS interface server to the listening client.

Error Handling	Reflected in HTTP response message.
HTTP Method	POST
URI	/SWO/StepNotification

XML schema:

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="SwitchingOrderStepNotification" type="tns:SwitchingOrderStepNotification" />
<xs:complexType name="SwitchingOrderStepNotification">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="WorkRequestId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="SwitchingOrderName" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Version" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="StepId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="StepNumber" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceID" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="StationName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceState" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="IssuedBy" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="IssuedTo" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="StepInstruction" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="IssuedTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="CompletedTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="ActualTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ModifiedByUser" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ModifiedTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="SwitchOrderNumber" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```


6. Switching Order WFM Web Service

The Switching Order WFM web service enables communication with an external workforce management system to exchange switching order data. The client URL is configured by the PluginApiConfig parameters.

The following calls are currently supported:

- Send a switching order
- Get switching order updates
- Get message processing status

6.1. Service API

This section provides details about service calls.

6.1.1. Send a Switching Order

DMS interface server uses this call to send a switching order to an external application.

HTTP Method	POST
URI	http://localhost:8443/api/ext/swo/v2/SwitchingOrder
Response	Successful request returns a Boolean value, True or False, which shows whether the request was processed. Any other response is considered an error.

XML request schema:

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="SwitchingOrder" nillable="true" type="SwitchingOrder" />
  <xs:complexType name="SwitchingOrder">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Name" nillable="true" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="WorkRequestId" nillable="true" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="ReferenceNumber" nillable="true" type="xs:string" />
    </xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="StationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="DeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="FeederName" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Orglevel2" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="OrgLevel1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Version" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="IsPlanned" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ScheduledStartTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ScheduledEndTime" type="xs:dateTime" />
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="0" maxOccurs="1" name="ActualStartTime" nillable="true"
type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="ActualEndTime" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="CreatedTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="LastModifiedTime" nillable="true"
type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="PreparedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ValidatedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ValidatedTime" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="ApprovedBy1" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ApprovedBy2" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ApprovedBy3" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ApprovedTime1" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="ApprovedTime2" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="ApprovedTime3" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="ApprovedTo" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ExecutedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CompletedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ExecutedTime" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="Description" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="SearchKeyWord" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastModifiedByUser" nillable="true"
type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Steps" nillable="true"
type="ArrayOfSwitchingOrderStep" />
        <xs:element minOccurs="1" maxOccurs="1" name="AuditTrails" nillable="true"
type="ArrayOfSwitchingOrderAuditTrail" />
        <xs:element minOccurs="1" maxOccurs="1" name="CustomFields" nillable="true"
type="ArrayOfCustomField" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfSwitchingOrderStep">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderStep" nillable="true"
type="SwitchingOrderStep" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="SwitchingOrderStep">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Id" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="StepNumber" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceID" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceName" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceTypeName" nillable="true" type="xs:string"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceStationName" nillable="true"
type="xs:string" />
    </xs:sequence>
</xs:complexType>

```

```

<xs:element minOccurs="1" maxOccurs="1" name="DeviceState" nillable="true" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="TagType" nillable="true" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="StepInstruction" nillable="true" type="xs:string"
/>
<xs:element minOccurs="1" maxOccurs="1" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="IssuedBy" nillable="true" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="IssuedTo" nillable="true" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IssuedTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="CompletedTime" nillable="true" type="xs:dateTime"
/>
<xs:element minOccurs="0" maxOccurs="1" name="ActualTime" nillable="true" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="CustomFields" nillable="true"
type="ArrayOfCustomField" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCustomField">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="CustomField" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="CustomField">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Name" nillable="true" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Value" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfSwitchingOrderAuditTrail">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderAuditTrail" nillable="true"
type="SwitchingOrderAuditTrail" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="SwitchingOrderAuditTrail">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="ModifiedbyUser" nillable="true" type="xs:string"
/>
<xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="DeviceState" nillable="true" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ModifiedTime" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="Description" nillable="true" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Version" type="xs:unsignedInt" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

6.1.2. Get Switching Order Updates

DMS interface server uses this call to get switching order updates from an external application.

HTTP Method	GET
-------------	-----

URI	http://localhost:8443/api/ext/swo/v2/SWOMessages
Response	SwitchOrderResponseV2
Frequency	Use the SwoMessagesInterval parameter to configure the time period for automatically sending these requests

XML response schema:

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="SwitchOrderResponseV2" nillable="true" type="SwitchOrderResponseV2" />
  <xs:complexType name="SwitchOrderResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="MaxResults" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpTo" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="SwitchingOrderUpdateMessages"
type="ArrayOfSwitchingOrderUpdateV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="SwitchingOrderFieldUpdateMessages"
type="ArrayOfSwitchingOrderFieldUpdateV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="SwitchingOrderFieldUpdateV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="WorkRequestId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FieldName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="NewValue" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="SwitchingOrderUpdateV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Number" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="WorkRequestId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Url" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Pdf" type="xs:base64Binary" />
                <xs:element minOccurs="0" maxOccurs="1" name="SwitchingOrder" type="SwitchingOrder" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="SwitchingOrder">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Name" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="WorkRequestId" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ReferenceNumber" nillable="true" type="xs:string" />
    </xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="StationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="DeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="FeederName" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Orglevel2" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="OrgLevel1" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Version" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="IsPlanned" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ScheduledStartTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ScheduledEndTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ActualStartTime" nillable="true"
type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ActualEndTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="CreatedTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="LastModifiedTime" nillable="true"
type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="PreparedBy" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ValidatedBy" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ValidatedTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="ApprovedBy1" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ApprovedBy2" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ApprovedBy3" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ApprovedTime1" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ApprovedTime2" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="ApprovedTime3" nillable="true" type="xs:dateTime" />

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="ApprovedTo" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ExecutedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CompletedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ExecutedTime" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="Description" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="SearchKeyWord" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastModifiedByUser" nillable="true"
type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Steps" nillable="true"
type="ArrayOfSwitchingOrderStep" />
        <xs:element minOccurs="1" maxOccurs="1" name="AuditTrails" nillable="true"
type="ArrayOfSwitchingOrderAuditTrail" />
        <xs:element minOccurs="1" maxOccurs="1" name="CustomFields" nillable="true"
type="ArrayOfCustomField" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfSwitchingOrderStep">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderStep" nillable="true"
type="SwitchingOrderStep" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="SwitchingOrderStep">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Id" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="StepNumber" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceID" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceName" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceTypeName" nillable="true" type="xs:string"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceStationName" nillable="true"
type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DeviceState" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="TagType" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="StepInstruction" nillable="true" type="xs:string"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="Comment" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="IssuedBy" nillable="true" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="IssuedTo" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="IssuedTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="CompletedTime" nillable="true" type="xs:dateTime"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="ActualTime" nillable="true" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="CustomFields" nillable="true"
type="ArrayOfCustomField" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCustomField">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="CustomField" />
    </xs:sequence>

```

```

</xs:complexType>
<xs:complexType name="CustomField">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Name" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfSwitchingOrderAuditTrail">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderAuditTrail" nillable="true"
type="SwitchingOrderAuditTrail" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="SwitchingOrderAuditTrail">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="ModifiedbyUser" nillable="true" type="xs:string"
/>
    <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="DeviceState" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ModifiedTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="Description" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Version" type="xs:unsignedInt" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfSwitchingOrderUpdateV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderUpdateV2" nillable="true"
type="SwitchingOrderUpdateV2" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfSwitchingOrderFieldUpdateV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="SwitchingOrderFieldUpdateV2"
nillable="true" type="SwitchingOrderFieldUpdateV2" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

6.1.3. Get Switching Message Processing Status

DMS interface server uses this call to get the message processing status from an external application.

HTTP Method	GET
URI	http://localhost:8443/api/ext/swo/v2/Status
Response	Status
Frequency	On demand

XML response schema:

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="Status" nillable="true" type="Status" />
  <xs:complexType name="Status">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="ProcessName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
      <xs:complexType mixed="true">
        <xs:sequence>
          <xs:any maxOccurs="unbounded" />
        </xs:sequence>
        <xs:anyAttribute />
      </xs:complexType>
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```


7. Annotation Web Service

The Annotation web service can be used to show data from external systems on the network view by creating, modifying, and deleting annotations. The web service is based on the DMS Control Service (see [DMS Control Service](#)). It can be accessed via the following URI:
<http://localhost:8801/dms/services/annotation>.

The currently supported external data types are Trucks and Outage Tickets.

To check for synchronization among an external system and the ADMS, the DMS interface server records the last updated time for annotation requests. If the connection between systems is lost for an extended period of time, the external system queries the DMS interface server for the last updated time. After that, the external system can re-send all changed (created, modified, deleted) annotations to the ADMS. If the DMS interface server restarts, it queries for all existing annotations to check the last updated time.

The following calls are currently supported:

- Create/update an annotation
- Get an annotation
- Delete an annotation
- Get the last update time

7.1. Service API

This section provides details about service calls. For more information, refer to <http://localhost:8801/dms/services/annotations/help>.

7.1.1. Create/Update Annotation

This type of call is used by an external entity to send a new annotation to the DMS interface server, or update the existing one.

When the DMS interface server receives a PUT/POST annotation call, it checks the annotation ID. If an annotation with the given ID already exists, the DMS interface server updates it. If the ID is unique, the DMS interface server creates a new annotation.

The API is limited in that the device id must be provided to associate the annotation with a device. When the device id and device type are supplied the devices link of the new annotation will be populated. When using the API to add annotations, the Status bitmap in the Annotation Summary display will not be populated automatically as this is a UI dependent feature. Subsequent editing and saving the annotation will populate the Status bitmap in Summary display.

This operation is asynchronous. If the request is successful, the annotation is stored on the DMS server available for modifying and deleting.

HTTP Method	PUT or POST
URI	/services/annotations/
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 500: Generic Server Error501 – Cannot change existing annotation type (for update only)
Notes	ID is used as a key for a Get/Delete call.

The annotation schema shows the required and optional properties:

```
<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:complexType name="Annotation">
  <xs:sequence>
    <xs:element name="Id" nillable="true" type="xs:string" />
    <xs:element name="Name" nillable="true" type="xs:string" />
    <xs:element name="Type" type="xs:int" />
    <xs:element name="Status" type="xs:int" />
    <xs:element name="Latitude" type="xs:double" />
    <xs:element name="Longitude" type="xs:double" />
    <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ServiceCenter" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DeviceType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CreationTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="UpdateTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="CreatedBy" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="UpdatedBy" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="Annotation" nillable="true" type="tns:Annotation" />
</xs:schema>
```

7.1.2. Get Annotation

This type of call is used to query an annotation by ID. This is a synchronous operation: the annotation detail is returned in the response payload if the operation is successful.

HTTP Method	Get
URI	/services/annotations/{Id}

Response	Success: • 200: Payload contains the annotation as defined in the above schema. Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error
----------	---

7.1.3. Delete Annotation

This type of call is used to delete an annotation by ID of the annotation. If the ID is not available, the system will return an error message.

HTTP Method	Delete
URI	/services/annotations/{Id}
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error

7.1.4. Get LastUpdateTime

DMS interface server maintains the last update time for all annotations and for specific types. The last update time can be used to determine if ADMS and the external system are out of sync.

If there are no annotations, the system returns DateTime.MinValue in C#, which is equivalent to "0001-01-01T00:00:00".

HTTP Method	Get
URI	/services/annotations/lastUpdateTime?Type={type}
Response	Success: • 200: DateTime Failure: • 401: Unauthorized • 500: Generic Server Error
Parameters	• Type for the annotation. If the type is not provided, the system returns the last update time for all existing annotation types.

8. Schedules Web Service

The Schedules web service can be used to send or update dynamic schedule data to ADMS. The web service is based on the DMS Control service (see [DMS Control Service](#)). It can be accessed via the following URI: `http://localhost:8801/dms/services/schedules`.

The currently supported external data type is Schedule.

The following calls are currently supported:

- Create/update a schedule for a device
- Get a schedule
- Create/update a schedule for a point
- Create/update a list of schedules

8.1. Service API

This section provides details about service calls. For more information, refer to `http://localhost:8801/dms/services/schedules/help`.

8.1.1. Create/Update Schedule for Device

This type of call is used by an external entity to send a new dynamic schedule to the DMS interface server, or update the existing one.

When the DMS interface server receives a PUT/POST schedule call, it sends the schedule to the DMS server. The DMS server updates the schedule data point if it already exists. If a point does not exist, the DMS server creates it.

This operation is asynchronous. If the request is successful, the schedule is stored on the DMS server.

HTTP Method	PUT or POST
URI	<code>/services/schedules/</code>
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 500: Generic Server Error501 – Cannot change existing schedule type (for update only)

The schedule schema shows the required and optional properties:

```
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
```

```

targetNamespace="http://www.esca.com/ids" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="Schedule">
    <xs:sequence>
      <xs:element name="DeviceId" nillable="true" type="xs:string" />
      <xs:element name="GroupId" nillable="true" type="xs:string" />
      <xs:element name="XUnit" type="xs:int" />
      <xs:element name="YUnit" type="xs:int" />
      <xs:element minOccurs="0" name="YUnit2" nillable="true" type="xs:int">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element minOccurs="0" name="YUnit3" nillable="true" type="xs:int">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element name="X" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfdateTime" />
      <xs:element name="Y" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfdouble" />
      <xs:element minOccurs="0" name="Y2" nillable="true"
xmlns:q3="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q3:ArrayOfdouble">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element minOccurs="0" name="Y3" nillable="true"
xmlns:q4="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q4:ArrayOfdouble">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element name="ScheduleType" type="xs:int" />
      <xs:element minOccurs="0" name="ScheduleId" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="Schedule" nillable="true" type="tns:Schedule" />
</xs:schema>

```

8.1.2. Get Schedule

HTTP Method	GET
URI	/services/schedules/?deviceId={DEVICEID}&groupId={GROUPID}&scheduleType={SCHEDULETYPE}
Response	Success: • 200: Payload contains the schedules as defined in the schema below Failure: • 401: Unauthorized • 404: Not found • 500: Generic Server Error

Schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="Schedule">
    <xs:sequence>
      <xs:element name="DeviceId" nillable="true" type="xs:string" />
      <xs:element name="GroupId" nillable="true" type="xs:string" />
      <xs:element name="XUnit" type="xs:int" />
      <xs:element name="YUnit" type="xs:int" />
      <xs:element minOccurs="0" name="YUnit2" nillable="true" type="xs:int">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          />
        />
      />
      <xs:element minOccurs="0" name="YUnit3" nillable="true" type="xs:int">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          />
        />
      />
      <xs:element name="X" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfdateTime" />
      <xs:element name="Y" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfdouble" />
      <xs:element minOccurs="0" name="Y2" nillable="true"
xmlns:q3="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q3:ArrayOfdouble">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          />
        />
      />
    />
  />
</xs:complexType>
```

```

        </xs:appinfo>
        </xs:annotation>
    </xs:element>
    <xs:element minOccurs="0" name="Y3" nillable="true"
xmlns:q4="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q4:ArrayOfdouble">
        <xs:annotation>
            <xs:appinfo>
                <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
            </xs:appinfo>
        </xs:annotation>
    </xs:element>
    <xs:element name="ScheduleType" type="xs:int" />
    <xs:element minOccurs="0" name="ScheduleId" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="Schedule" nillable="true" type="tns:Schedule" />
</xs:schema>

```

8.1.3. Create/Update Schedule for Point

HTTP Method	PUT or POST
URI	/services/schedules/1
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 500: Generic Server Error501 – Cannot change existing schedule type (for update only)

Schema:

```

<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
    <xs:complexType name="ScheduleOfPoint">
        <xs:sequence>
            <xs:element name="X" type="xs:dateTime" />
            <xs:element name="ScheduleId" nillable="true" type="xs:string" />
            <xs:element name="Data" nillable="true" type="tns:ArrayOfScheduleOfGroup" />
        </xs:sequence>
    </xs:complexType>
    <xs:element name="ScheduleOfPoint" nillable="true" type="tns:ScheduleOfPoint" />
    <xs:complexType name="ArrayOfScheduleOfGroup">
        <xs:sequence>
            <xs:element minOccurs="0" maxOccurs="unbounded" name="ScheduleOfGroup" nillable="true"
type="tns:ScheduleOfGroup" />
        </xs:sequence>
    </xs:complexType>
    <xs:element name="ArrayOfScheduleOfGroup" nillable="true" type="tns:ArrayOfScheduleOfGroup" />

```

```

<xs:complexType name="ScheduleOfGroup">
  <xs:sequence>
    <xs:element name="GroupId" nillable="true" type="xs:string" />
    <xs:element name="Data" nillable="true" type="tns:ArrayOfScheduleOfDevice" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ScheduleOfGroup" nillable="true" type="tns:ScheduleOfGroup" />
<xs:complexType name="ArrayOfScheduleOfDevice">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="ScheduleOfDevice" nillable="true"
type="tns:ScheduleOfDevice" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfScheduleOfDevice" nillable="true" type="tns:ArrayOfScheduleOfDevice" />
<xs:complexType name="ScheduleOfDevice">
  <xs:sequence>
    <xs:element name="DeviceId" nillable="true" type="xs:string" />
    <xs:element name="Data" nillable="true" type="tns:ArrayOfScheduleData" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ScheduleOfDevice" nillable="true" type="tns:ScheduleOfDevice" />
<xs:complexType name="ArrayOfScheduleData">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="ScheduleData" nillable="true"
type="tns:ScheduleData" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfScheduleData" nillable="true" type="tns:ArrayOfScheduleData" />
<xs:complexType name="ScheduleData">
  <xs:sequence>
    <xs:element name="XUnit" type="xs:short" />
    <xs:element name="YUnit" type="xs:short" />
    <xs:element name="ScheduleType" type="xs:short" />
    <xs:element name="Y" type="xs:double" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ScheduleData" nillable="true" type="tns:ScheduleData" />
</xs:schema>

```

8.1.4. Create/Update List of Schedules

HTTP Method	PUT or POST
URI	/services/schedules/Set
Response	Success: • 200: No payload Failure: • 401: Unauthorized • 500: Generic Server Error501 – Cannot change existing schedule type (for update only)

Schema:

```
<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="ArrayOfSchedule">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="Schedule" nillable="true"
type="tns:Schedule" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfSchedule" nillable="true" type="tns:ArrayOfSchedule" />
  <xs:complexType name="Schedule">
    <xs:sequence>
      <xs:element name="DeviceId" nillable="true" type="xs:string" />
      <xs:element name="GroupId" nillable="true" type="xs:string" />
      <xs:element name="XUnit" type="xs:int" />
      <xs:element name="YUnit" type="xs:int" />
      <xs:element minOccurs="0" name="YUnit2" nillable="true" type="xs:int">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element minOccurs="0" name="YUnit3" nillable="true" type="xs:int">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element name="X" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfdateTime" />
      <xs:element name="Y" nillable="true"
xmlns:q2="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q2:ArrayOfdouble" />
      <xs:element minOccurs="0" name="Y2" nillable="true"
xmlns:q3="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q3:ArrayOfdouble">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
      <xs:element minOccurs="0" name="Y3" nillable="true"
xmlns:q4="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q4:ArrayOfdouble">
        <xs:annotation>
          <xs:appinfo>
            <DefaultValue EmitDefaultValue="false"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/" />
          </xs:appinfo>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>

```

```
        </xs:appinfo>
        </xs:annotation>
    </xs:element>
    <xs:element name="ScheduleType" type="xs:int" />
    <xs:element minOccurs="0" name="ScheduleId" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:element name="Schedule" nillable="true" type="tns:Schedule" />
</xs:schema>
```

9. DMS Interface Server Configuration

DMS interface server reads two configuration files:

- **netconf_dataservice.xml**, which is used to configure the connection to the DMS server (NetSrv).
- **DmsInterfaceServerConfig.xml**, which is used to configure the DMS interface server itself.

To start DMS interface server, you can use a shortcut similar to the following:

```
> D:\eterra\Distribution\Server\bin\DmsInterfaceServer.exe /CONFIG  
%IDMS_ROOT%\config\client\netconf_dataservice.xml  
%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml
```

9.1. netconf_dataservice.xml

For descriptions of the netconf_dataservice.xml file parameters, refer to the *Net Dynamics Configuration Editor User Guide*.

In the netconf_dataservice.xml file, the following settings need to be updated during deployment:

- **CustomSecretStore:** The DMS Interface Server can retrieve the DMS Interface Server trusted user ID from the CyberArk Enterprise Password Vault so that you do not need to provide the value in the configuration file (as a plain text or encrypted string). For more information, refer to the *ADMS Software Installation and Maintenance Guide*.
 - **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.
 - **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
 - **ObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the DMS Interface Server trusted user ID defined in the Enterprise Vault Server. The format is
<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.
- **LogFile:** This setting configures the location and name of the DMS interface server log file.
- **AnnotationData:** These settings configure the types of annotation data that will be supported in the system.
 - **Annotation Type:** Sequential numeric value of the annotation.
 - **Annotation Name:** Annotation name.
 - **Annotation Mode:** Annotation mode (e.g., journal).
 - **Fixed:** True or False. Fixed annotations can only be moved by users who have specific

permissions.

- **Annotation File:** <new annotation name>.sav.

The annotation dynamics will be recorded to this file.

- **SafetyRUserCommand:** RUSER command to move the safety document on the Geographic Network view.
- **StudyServerModelPath:** To allow for the automated features of the CopyFromRealTime and savecase capabilities, the client needs to know about the Study server's model path. This should be the same as the folder defined in the Study server's configuration file.
- **StudyServerDynamicsPath:** To allow for the automated features of the CopyFromRealTime and savecase capabilities, the client needs to know about the Study server's dynamics path. This should be the same as the folder defined in the Study server's configuration file.
- **SaveCasePath:** Set to the folder where savecase information should be stored.
- **CRTPath:** Path to the CopyFromRealTime (read only) folder.
- **CopyFromRTAdjacency:** Specified as an integer, this number defines the level of adjacency when the stations are copied from the real-time server. To enable the retrieval of all station model files, set this parameter to a number greater than the total number of model files. For example, if there are 52 stations, set this value to 60. This will retrieve all of the model files in the folder regardless of geographical or electrical adjacency and account for future model additions.
- **SubscribeForGisNotifications:** Specifies whether the DMS interface server receives from the DMS server notifications about switch status changes to the Pending/Permanent state.
- **ApplicationType:** The DMS interface server application type. The default value is "DIS".

The application type is required to distinguish the DMS interface server from a client when monitoring the connection and synchronization with the enabled server.

Note

To issue an alarm for DMS interface server connection and synchronization status changes, add "DIS" to the MONITOR_CONNECTION_STATUS_OF_CLIENTS and MONITOR_DYN_SYCH_STATUS_OF_CLIENTS parameters in the main server configuration file.

- **OmsClientConfigName:** Sets the full path to the OMS client configuration file.
- **CodedTransConfigName:** Sets the path to the coded translation configuration file.
- **CustomerFileName:** Sets the full path to the OMS customer file.
- **Timeout:** This configures how long the dynamic query will wait for the result to come back from the DMS server. The default value is 10 seconds. If the wait exceeds the timeout, an error message is returned indicating that the query failed because of a timeout from the server.
- **ViewerConfigName:** Name of the viewer configuration.
- **Port:** Specifies the port number that the DMS server is listening on. For different types of

servers, it has different default ports; for example, RT uses 5025 and ST uses 5027.

- **ServerIPs:** Configures the NetSrv IP address. It allows up to four server addresses in a comma-delimited list. If multiple servers are configured, the DMS interface server tries to connect to each server in a round-robin manner for redundancy.
- **ModelPath:** Sets the path to the station model files.
- **Mode:** The default value is ClientMode.
- **Name:** Mode name. The default value is RT.

9.2. DmsInterfaceServerConfig.xml

In `DmsInterfaceServerConfig.xml`, you can modify the following settings based on deployment needs.

- **EnableDMSDataServices:** By default, this is set to true. Setting `EnableDMSDataServices` to false disables the DMS Data service.
- **EnableDMSControlServices:** By default, this is set to true. Setting `EnableDMSControlServices` to false disables the DMS Control service. This option must be set to true to enable the Annotation web service.
- **EnableSwitchingOrderWebServices:** By default, this is set to true. Set `EnableSwitchingOrderWebServices` to false to disable the Switching Order web service.
- **EnableMobileWebServices:** By default, this is set to false. Set `EnableMobileWebServices` to true to enable the Switching Order Mobile web service.
- **EnableWfmServices:** By default, this is set to false. Set `EnableWfmServices` to true to enable the Switching Order WFM web interface for exchanging switching order data with WFM.
- **EnableDerControlServices:** By default, this is set to false. Set `EnableDerControlServices` to true to enable the DER Control web service and poll the DNAS server for Distributed Energy Resource (DER) control data for exchanging data with the 2030.5 Gateway.
- **EnableGisNotificationHandler:** By default, this is set to false. Set `EnableGisNotificationHandler` to true to enable forwarding notifications about switch status changes to the Pending/Permanent state to a remote endpoint specified by the `GisNotificationDispatchUrl` parameter.
- **ReadOnlyDMS:** By default, this is set to false. In most cases, DMS interface server will connect to NetSrv. Setting `ReadOnlyDMS` to true disables most of the server functionality. Most of the dynamic data will not be available and the server will automatically disable those queries, as well as archiving.
- **EnableFAService:** By default, this is set to true. Set `EnableFAService` to false to disable the Forecast Adapter web service.
- **FAFlatfileContent:** If true, the Forecast Adapter web service output is shown in a comma-delimited flat file format. If false, the Forecast Adapter web service output is shown in a structured XML file format.

- **FAInterestedStations:** The list of substation names (separated by commas) that defines the substations the Forecast Adapter web service clients are interested in. For substations that are not included in the list, the first client's query for data takes longer to resolve because the Forecast Adapter service waits for a result from the DNAS Server.
- **QueryIntervalInSeconds:** Configures how often the DMS interface server queries the DMS server for data updates. The default value is 600 seconds.
- **KeepWSDataInDays:** Configures how long the DMS interface server will keep certain DMS data in memory for web service queries (including DSO solutions and Fault Location solutions). The default value is one day.
- **UseUAAForGettingToken:** Specifies whether UAA authentication is enabled. Enables retrieving a valid UAA authentication token from the UAA server.
- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **UseUAATokenForIntrospect:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token. The DMS Interface Server extracts a UAA authentication token from the authorization header of incoming requests and sends the token to the UAA server for validation.
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **WebServiceBaseURL:** Configures the base URL for trusted applications. Note that the trailing "/" is needed at the end. The default value is `http://localhost:8801/dms/`.

Each web service application adds a specific ending to this base. For example, the URL base for the Switching Order web service is `http://localhost:8801/dms/SWO`.

- **SecureWebService:** Configures whether a secure web service connection is used. The default value is false.
- **CertSubjectName:** Configures Certificate client Subject (CN) in mutual tls connection. If SecureWebService is true and CertSubjectName is empty then Server Side only TLS is enabled.
- **RevocationMode:** Configures the certificate revocation mode for client in mutual tls connection (NoCheck 0, Online 1, Offline 2).
- **DisableAllArchive:** Configures whether all queries for archiving purposes are disabled. The default value is false.

Note

Archiving options can be changed at run time on the DMS interface server display from the Archive Options menu.

- **DisableFISRArchive:** Configures whether querying for FISR solution data for archiving purposes is disabled. The default value is false.
- **DisableVVCArchive:** Configures whether querying for LVM solution data for archiving purposes is disabled. The default value is false.
- **DisableAFRArchive:** Configures whether querying for AFR solution data for archiving purposes

is disabled. The default value is false.

- **DisableFLArchive:** Configures whether querying for FL solution data for archiving purposes is disabled. The default value is false.
- **DisableDPFArchive:** Configures whether querying for DPF solution data for archiving purposes is disabled. The default value is false.
- **DisableStationFilesArchive:** Configures whether querying for station file data for archiving purposes is disabled. The default value is false.
- **DisableStationExceptionArchive:** Configures whether querying for station exception data for archiving purposes is disabled. The default value is false.
- **DisableFeederExceptionArchive:** Configures whether querying for feeder exception data for archiving purposes is disabled. The default value is false.
- **DisableAllDpfSubQueries:** If set to true, all DPF-related queries configured by the `DisableDPFOutputDataQueries`, `DPFTransformerLossResults`, `DpfMeasurementIslandQuery`, `DpfFeederLossesQuery`, `FeederExceptionQuery`, and `StationExceptionQuery` parameters are disabled when DPF is solved for a substation.

 Note

DPF query options can be changed at run time on the DMS interface server display from the Query Options menu.

- **DisableDPFOutputDataQueries:** If set to false, the DMS interface server querying for the voltage output, voltage violation, and overload violation data when DPF solves is enabled. This option is set to true by default to reduce the amount of data sent. This option is required only for the DERMS project.
- **DisableDPFTfLossResultsQuery:** Configures whether querying for DPF transformer loss results is disabled.
- **DisableDPFMeasIslandQuery:** Configures whether querying for DPF/BLA measurement island results is disabled.
- **DisableDPFFeederLossQuery:** Configures whether querying for DPF feeder loss results is disabled.
- **DisableFeederExceptionQuery:** Configures whether querying for DPF feeder exceptions results is disabled.
- **DisableStationExceptionQuery:** Configures whether querying for DPF station exception results is disabled.
- **DisableServiceAuthCheck:** If this is set to true, the authentication check for control service calls is disabled. This is added for ease of testing and should **not** be set to true in a production system.
- **EnableMessageLog:** Configures whether the DMS interface server logs all the incoming web service requests and output response. The default value is false.
- **LogLevel:** An integer value that controls the types of log messages written to the log. Higher

values result in more detailed log messages. The values are: 1=InfoAlways, 2=Fatal, 3=Error, 4=Warning, 5=Info, 6-8=Debug.

- **LogFile:** This configures the log file location, for example %IDMS_ROOT%\log\DmsInterfaceServer.txt.
- **MaxMessagesInLogFile:** Configures the maximum number of messages in the log file before the system creates a new one. The default value is 10000.
- **ArchiveFilePath:** Configures the path for the DMS interface server to save its archive journal files.
- **MaxMessagesInArchiveFile:** Configures the maximum number of messages in the journal file before the system creates a new one. The default value is 10000.
- **ConverterDLL:** Path to the external DLL location. The default value is NetUtil.dll.
- **ConverterClassName:** Assembly and type information in the format AssemblyName.TypeName. For example: SampleCoordinateConverter.CoordinateConverter.
- **CustomConverterOptions:** Parameters of a converter that converts between projected coordinate system (PCS) coordinates and system coordinates. Coordinate conversion is required for annotation web service calls.

For US customers, the default coordinate converter can be used. For non-US customers, a custom converter must be developed.

- **ConverterDLL:** Name of the assembly (DLL) that contains a class for converting between latitude/longitude and system X/Y coordinates.
- **ConverterClassName:** Name of the class that converts between latitude/longitude and system X/Y coordinates.
- **GeographicName:** Sets the name of the geographic geometry that is used in the display call-up URLs. The default name is DETAIL.
- **EnableTrustedClientDualSSL:** To turn on Dual ssl, set EnableTrustedClientDualSSL to true and add the TrustedClientCertStore and TrustedClientCertSubjectName. The default value is false.
- **TrustedClientCertSubjectName:** Configures trusted client certificate subject (CN).
- **TrustedClientCertStore:** Configures which trusted client certificate store to use. The default value is Personal.
- **TrustedClientList and TrustedClient:** These elements are used to set up trusted application lists. Each application can have the following settings:
 - **Name:** Used to identify the application. For example, Notification Publisher, DnafNotificationPublisher, ForecastAdapter, or Tag Event Publisher.
 - **Key:** Used to calculate the hash value for the authorization header. Required if DisableServiceAuthCheck is set to false. ***NotificationBaseURL:** The base URL for trusted applications to receive notifications. The default value is http://localhost:9090/dms/event/.
 - For Notification Publisher, use http://localhost:8025/DISEvents/.
 - For DnafNotificationPublisher, use http://localhost:8901/DISEvents/.

- For Forecast Adapter, use `http://<DockerHost>:8078/api/v1/dms/`.
- For the Mobile server application, use `http://mobilityserver:9180/event/`.
- For Tag Event Publisher, use `http://localhost:8443/api/adms/MobileSwo/`.

The DMS interface server uses this base URL as an address where the POST web request for the notification will be sent. For example, the POST web request for the Switching Order status notification will be sent to `http://localhost:9090/dms/event/SWO/StatusNotification`.

- **NotificationTypes:** The notification types that the application is interested in. The valid values are SWITCH, TAG, DNAF, SWO-ALL, SWO-STATUS, SWO-STEP, SWO-PDF, and SWO-Status-EXT. If not specified, the default value is ALL. For Notification Publisher, use SWO-Status-EXT. For DnafNotificationPublisher, use DNAF. For Forecast Adapter, use SWITCH.

For a "TAG" notification type, the string "tagStatus" will be appended to the NotificationBaseUrl. For example, "NotificationBaseUrl/tagStatus"

- **NotificationFormat:** The format of the notification. The valid values are XML and JSON.
- **EnableDualSSL:** Configures the specific Trusted Client to be enabled for mutual tls. The default value is false.
- **SwitchingOrderServerCommConfig:** The Switching Order Server Communication Client is responsible for implementing the communication interface with the Switch Order server. It has the following options:
 - **ListenerHost:** Name or IP address of the host machine listening for client connections. The default value is localhost.
 - **ListenerPort:** TCP/IP port number on which the listener host is listening for client connections. The default value is 5044.
 - **TimeToLive:** Number of seconds before received messages are considered expired. The default value is 300 seconds.
 - **CompressMessage:** If this option is set to True, Switching Order Server Communication Client messages are compressed before sending. This decreases message size and improves performance. This CompressMessage option must be set with the same value as the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
 - **EnableTcpipTls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
 - **HelloIntervalSeconds:** Interval between periodic hello messages to check that the connection is still open. The default value is 30 seconds.
- **MobileServicesSettings:** The MobileServicesSettings element specifies the Mobile Switching Order web service settings:
 - **Deprecated_EnableXMLForMobileAdapter:** If set to false, JSON serialization and deserialization is used. This is required to integrate with the Mobile Enterprise system using the Mobile Switching Order web service (default functionality). If set to true, XML serialization and deserialization is enabled. This is required to integrate with the Mobile

Enterprise system using the Mobile Adapter (deprecated functionality).

- **MobileUrl:** URL of the Mobile Switching Order web service of the Communications Gateway application.
- **PollingDelayInMs:** Time in minutes before requesting new messages from the Mobile client application.
- **HealthStatusIntervalInMs:** Time in minutes before DIS checks the health status of the mobile web server.
- **UseUAAForGettingToken:** Specifies whether UAA authentication is enabled. Enables retrieving a valid UAA authentication token from the UAA server.
- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **TagRestSettings:** The TagRestSettings element specifies the Tagrest web service settings:
 - **Url:** Specifies URL of the Tagrest web service
 - **PermitUserName:** Specifies the username to be used in Tagrest messages in principal_user field.
- **DerControlSettings:** The DerControlSettings element specifies the DER Control web service settings:
 - **DermsGatewayDispatchUrl:** Configures the 2030.5 Gateway API URL, which the DER Control web service uses for exchanging DER data.
 - **PollingDelayInMs:** Time in minutes before requesting DER control updates from the DNAF server.
 - **UseUAAForGettingToken:** Specifies whether UAA authentication is enabled. Enables retrieving a valid UAA authentication token from the UAA server.
 - **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **GisNotificationControllerSettings:** The GisNotificationControllerSettings element specifies settings for the GIS notifications:
 - **GisNotificationDispatchUrl:** Configures the URL where the DMS Interface Server sends notifications about switch status changes to the Pending/Permanent state.
 - **SendRetryIntervalSeconds:** Time period in seconds for trying to resend a notification if the remote host is unavailable.
 - **JournalPath:** Path to the folder where GIS notification journals are recorded.
 - **JournalName:** Name for GIS notification journal files.
 - **MaximumJournalEntries:** The maximum number of entries to be written to a GIS notification journal. After the maximum number is reached, another journal file is created.
 - **JournalExpirationPeriod:** The number of days before GIS notification journals expire. Expired journals are not loaded.
- **WFM Switching Order Interface Web Service Client Plugin Parameters:** This group includes

WFM Switching Order web service client plugin parameters.

- **Enabled:** If set to true, the plugin is enabled.
- **Type:** The type that implements the plugin.
- **Name:** Plugin name.
- **BaseUri:** The web service root URL.
- **Name:** Plugin name to show on the user interface.
- **MessagesUri:** URL for SWOMessages messages.
- **SwitchingOrderUri:** URL for SwitchingOrder messages.
- **StatusUri:** URL for Status messages.
- **SwoMessagesInterval:** Polling time for SWOMessages messages, in seconds.
- **PostingRetryInterval:** Time period, in seconds, to retry posting messages to the remote endpoint when posting fails (for example, due to connection issues).
- **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
- **UseHeaderAuthorization:** If set to True, "authorization" header checking is enabled for incoming web service calls.
- **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSerialNumber:** The certificate's serial number.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **UseUAAForGettingToken:** Specifies whether UAA authentication is enabled. Enables retrieving a valid UAA authentication token from the UAA server.
- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.

10. Running DMS Interface Server as a Windows Service

The DMS interface server can run as a Windows service. This may be useful for an environment where the DMS interface server is running separately from other ADMS servers (for example, for security issues) and is not managed by the Process Starter. When running as a Windows service, the DMS interface server can be automatically started or stopped when the hosting machine starts or stops.

Note

The DMS interface server must be run by a user with administrator privileges.

To install the DMS interface server service:

1. Add the /INSTALL option to the server's shortcut command.

```
/CONFIG "%IDMS_ROOT%\config\client\netconf_dataservice.xml"  
"%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml" /INSTALL
```

2. Run the DMS interface server using the shortcut.

This installs the DMS interface server service and adds DMSInterfaceServerService to the Windows services list.

Alternatively, you can run the following command at a command line:

```
DmsInterfaceServer.exe /CONFIG "%IDMS_ROOT%\config\client\netconf_dataservice.xml"  
"%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml" /INSTALL
```

To start the DMS interface server service, do one of the following:

- From the Windows Start menu, navigate to Control Panel> Administrative Tools> Services> DMSInterfaceServerService> Start.
- Run the following command at a command prompt:

```
net start dmsinterfaceserverservice
```

To stop the DMS interface server service, do one of the following:

- From the Windows Start menu, navigate to Control Panel> Administrative Tools> Services> DMSInterfaceServerService> Stop.
- Run the following command at a command prompt:

```
net stop dmsinterfaceserverservice
```

To uninstall the DMS interface server service:

1. Add the /UNINSTALL option to the server's shortcut command.

```
/CONFIG "%IDMS_ROOT%\config\client\netconf_dataservice.xml"  
"%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml" /UNINSTALL
```

2. Run the server using the shortcut.

This uninstalls the DMS interface server service and removes DMSInterfaceServerService from the Windows services list.

Alternatively, you can run the following command from command line:

```
DmsInterfaceServer.exe /CONFIG "%IDMS_ROOT%\config\client\netconf_dataservice.xml"  
"%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml" /UNINSTALL
```

11. Structure Server Configuration

The Structure Server uses the `EquipmentAndStructure.index` file to return requested information about structures and equipment, and the association between them. Note that the Structure Server is a different process, not another interface hosted by the DMS Interface Server.

This chapter describes how to configure the Structure Server.

11.1. Prerequisites

- Microsoft .NET Framework 5 must be installed. Refer to "Prerequisites for Installing the ADMS Interfaces" in the *ADMS Software Installation and Maintenance Guide*.
- Certificate must be installed. Refer to "Importing Certificates to the ADMS Hosts" in the *Internal Notifications Software Installation and Maintenance Guide*.
- `EquipmentAndStructure.index` file must be generated. Refer to "Installing the Interfaces Software" in the *ADMS Software Installation and Maintenance Guide*.

11.2. Setup

Open the `StructureServerConfig.xml` file (`%IDMS_ROOT%\config\interfaces`). Make sure that the `CertificateSubject` matches the installed certificate.

11.3. Running the Server

As a user with administrator privileges, open a command window, navigate to `<installation path>\Interfaces\bincore`, and run `StructureServer.exe`. You should see output similar to the following:

```
C:\Program Files\GE\ADMS\Interfaces\bincore>StructureServer.exe
Unable to open variable file: Version_Variables.txt
Hosting environment: Production
Content root path: C:\Program Files\GE\ADMS\Interfaces\bincore
Now listening on: https://192.168.0.11:17642
Application started. Press Ctrl+C to shut down.
```

11.4. API

- `https://<server>:17642/api/structureids/{structureId}/equipment`
Retrieves the equipment on the specified structure ID.
- `https://<server>:17642/api/structureids/{structureId}/info`

Retrieves structure information for the specified structure ID.

- `https://<server>:17642/api/equipmentinformation/{equipmentId}/structures`

Retrieves structure IDs on the specified equipment ID.

- `https://<server>:17642/api/equipmentinformation/{equipmentId}/info`

Retrieves equipment information for the specified equipment ID.

11.5. Distributed Configuration

11.5.1. Interfaces Machine

1. Generate a certificate for the server.
2. Edit the `StructureServerConfig.xml` file, set `WebServiceBaseUrl` to the server IP address, and set `CertificateSubject` to the generated certificate.
3. Generate the `EquipmentAndStructure.index` file.

11.5.2. Client Machine

1. Copy the certificate to the client machine and import it to Trusted Root Certification Authorities.
2. Edit the `netconf_localhost.xml` file. Update `ServerIPs` for RT to the server machine.
3. Using the Viewer Configuration Editor, select the Enable Structure Information option.